

On the Universality of Comparability Grids for Measurement-Based Quantum Computation

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In partial fulfillment of the requirements
for the Mathematics Honors Program

May 2026

Abstract

The measurement-based quantum computing (MBQC) model presents a way to do quantum computations in constant depth by measuring a pre-prepared graph state. As such, there is interest in determining which graph states are universal for MBQC. The universality of the comparability grid is currently an open problem, and of some interest as if it is universal, so are all families of graph states with unbounded rank-width. In this thesis, we report progress on this problem by discussing a variety of approaches for potentially proving and disproving the universality of the comparability grid for quantum computation.

Acknowledgements

First, I would like to thank Professor Daniel Grier for introducing me to the weird and wonderful world of quantum computing. His advice has been invaluable these past two years. He has been an incredibly kind mentor and I owe much of my growth, academically and personally, to him.

I would also like to extend this gratitude to Du (Dan) Van, the primary collaborator of this project, who has always been available to listen to my rambles. I wish him the best of luck in his future studies at NUS. I would also like to thank Jackson Morris, who has always been available to mentor and provide feedback for this project. I am also thankful for Kunal Marwaha, who laid some of the groundwork for this project, and kindly advised us during his visit to UCSD.

I also thank my family and friends for their utmost support in all aspects of my academic journey. It is through their efforts that I may focus on my studies and pursue my aspirations without distraction.

Preface to the Second Rough Draft

Incomplete sections and comments are written in blue throughout.

One of the great difficulties writing the rough draft was figuring out where to stop. In other words, how much can I assume from the reader? To answer this question, the past Mathematics honors theses of Ray Tsai, Clara Woods, and Jared Hughes in particular have been most helpful; the first for its recency and treatment of graph theory, the second for its experimental and quantum focus, and the third for its approach to 3XOR and theoretical computer science. These three, among other theses, appear to take a very introductory stance, writing for a non-expert audience familiar with only the typical mathematical fields (i.e. a senior mathematics undergraduate).

Because of this, I have prepared a (perhaps excessively) extensive preliminary section and appendix. Furthermore, I may have overextended the thesis' scope, resulting in many incomplete sections in the rough draft. I was originally planning to cut out the preliminary section entirely, or perhaps place it some extended appendix. However, no prior honors thesis (to the best of my skimming) has ever used the appendix in such a manner, nor excluded so much introduction. Thus, since it is easier to cut than to add, I have left it all in.

Furthermore, the main results of the research connected to this thesis is still a work in progress — indeed, we have not resolved the main question! Because of this, I found it challenging to create some sort of cohesive presentation of the work besides an organized diary of failed attempts. Because of this, I recognize much work needs to be done to better “carve out” the original research from what is known to the field. At the same time, I feel there is merit in surveying all the developments pertaining this study, even if this dwarfs the original contributions.

DRAFT 2

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1 Introduction

We start with an informal introduction to quantum computing and lay some motivations.

Fault-tolerant quantum computing has the potential, through algorithms like Shor’s factoring algorithm [Sho97], to deliver possibly exponential speedups over classical computation. However, current Noisy Intermediate-Scale Quantum (NISQ) computers have too few qubits and too much noise to deliver practical benefit [Pre18].

Computation in the NISQ era is often modelled by a hybrid setup: on the quantum computer, a noisy initial state is prepared, some noisy gates are applied, and qubits are measured. A classical computer then processes these measurements, and can modify and repeat the quantum computation if necessary.

For the quantum computer, there exist a variety of computational models: The quantum Turing machine [Deu85], the quantum circuit model [Bar+95], adiabatic quantum computation [Far+01], and measurement-based models [RB01], among many others. Though these models are equivalent from a complexity viewpoint (informally, any problem that is efficiently solvable in one is efficiently solvable in the others) [RB01; Aha+08; Chi93], the underlying quantum properties used by each may differ significantly.

The measurement-based quantum computation (MBQC) model is of particular interest. This is because MBQC collapses any quantum circuit into measurements of a highly-entangled resource state that may be prepared in constant depth. This is quite shocking, as it means that any quantum computation may be done in constant depth. The cost of this model, however, is that measurements need to be adaptive. In other words, processing must be done on-the-fly to determine what qubits should be measured next and how.

Since measurement is irreversible, MBQC leads to a model of one-way quantum computation. In other words, as measurement cannot increase the entanglement of a system, all entanglement required must be initially present in the resource state. Informally then, entanglement is the resource consumed by this model of quantum computation. Thus, there is interest in determining what resource states are universal for quantum computation.

Graph states are a kind of resource state whose data corresponds to a simple, undirected graph. The comparability grid is a graph with vertices $V = \{(x, y) \mid x, y \in [n]\}$ and an edge between (x_1, y_1) and (x_2, y_2) if $x_1 \geq x_2$ and $y_1 \geq y_2$, excluding self-loops. It is an open problem whether the family of all comparability grids is universal for MBQC. There are implications for both possibilities. For example, if comparability grids are not universal, then unbounded rank width is not a sufficient condition for the universality of a family of graph states, since the family of all comparability grids have unbounded rank width.

In this thesis, we report on progress to prove and disprove the universality of comparability grids. Concretely, we give a set of conjectures for how one may be able to extract universal quantum computation using comparability grids as our resource state. In tandem to this, we outline various algorithms to verify small cases of these conjectures.

Readers with no prior exposure to quantum computation are encouraged to start at Appendix A. In Section 2, we introduce the necessary notation and background so that in Section 3 we may formally define the concepts in our problem statement. Then, Section 4 recalls some results regarding comparability grids. These theoretical results inspire the algorithmic approaches in Section 5, which in turn guide the results in Section 6.

2 Background

First some notation. Let $\mathbb{N} = \{1, 2, \dots\}$. For $k \in \mathbb{N}$, define $[k] := \{1, \dots, k\}$ and $[0] := \emptyset$. Let I be the identity matrix of any size and of any vector space, which will be clear from context. For a graph G , let $V(G)$ denote its vertex set and $E(G)$ its edge set. For any vector v with n entries, let v_i be its i th entry for all $i \in [n]$. For any matrix A with n rows and m columns, let $A_{i,j}$ be the entry at the i th row and j th column, for all $i \in [n], j \in [m]$.

We assume the reader is familiar with the mathematical foundations of quantum computing (e.g. at the level of an upper undergraduate quantum computing class). Please see Appendix A otherwise.

2.1 The ZX-Calculus

DEFINITION 2.1 The **ZX-Calculus** is a graphical language for representing quantum computation. Objects of the ZX-Calculus, called **ZX Diagrams**, are made up of two types of generators known as **spiders**: The **X-spider**, with n inputs and m outputs, is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |0\rangle^{\otimes m} \langle 0|^{\otimes n} + e^{i\alpha} |1\rangle^{\otimes m} \langle 1|^{\otimes n}. \quad (2.1)$$

We say that the above spider has a **phase** $\alpha \in [0, 2\pi)$. Similarly, the **Z-spider** is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |+\rangle^{\otimes m} \langle +|^{\otimes n} + e^{i\alpha} |-\rangle^{\otimes m} \langle -|^{\otimes n}. \quad (2.2)$$

In other words, the ZX-Calculus is a calculus (i.e. rules) for manipulating Z and X spiders.

REMARK 2.2 In the literature, the X-spider is often red and the Z-spider is often green, e.g. ● and ●. However, for accessibility reasons we choose to adopt the black-and-white ZX-Calculus color scheme instead.

DEFINITION 2.3 Write $a \propto b$ if $a = c \cdot b$, for some $c \in \mathbb{C}$. Furthermore, if D is a ZX diagram, write $\llbracket D \rrbracket$ to denote the matrix represented by D . To reduce notational clutter, we will use $\llbracket D \rrbracket$ only when it is necessary to distinguish a diagram from its matrix.

EXAMPLE 2.4 The standard quantum states can be written as spiders without inputs:

$$\textcircled{0} \text{ --- } = \sqrt{2} |+\rangle \propto |+\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |-\rangle \propto |-\rangle, \quad (2.3)$$

$$\textcircled{0} \text{ --- } = \sqrt{2} |0\rangle \propto |0\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |1\rangle \propto |1\rangle. \quad (2.4)$$

Single qubit gates can be written as spiders with 1 input and 1 output:

$$\text{---} \textcircled{\alpha} \text{ ---} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix} = e^{i\alpha/2} R_z(\alpha) \quad (2.5)$$

$$\text{---} \textcircled{\alpha} \text{ ---} = \frac{1}{2} \begin{bmatrix} 1 + e^{i\alpha} & 1 - e^{i\alpha} \\ 1 - e^{i\alpha} & 1 + e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} R_x(\alpha) \quad (2.6)$$

Thus $\text{---} \boxed{Z} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$ and $\text{---} \boxed{X} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$.

DEFINITION 2.5 Informally, the composition of spiders in ZX diagrams is similar to the composition of quantum gates in quantum circuits: Spiders connected via wires multiply, while spiders “stacked vertically” are tensored together.¹

DEFINITION 2.6 Here is an example of ZX composition: The Hadamard gate satisfies

$$\boxed{H} = e^{-i\pi/4} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---}. \quad (2.7)$$

Due to this verbose representation, we shorthand $\text{---} \square \text{---} := \boxed{H}$ and call this the **H-box**.

DEFINITION 2.7 **Wires** or lines between spiders may “bend” in the following ways:

$$\text{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{C} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \text{) = } [1 \quad 0 \quad 0 \quad 1]. \quad (2.8)$$

These are called “swap,” “cup,” and “cap,” respectively from left to right, and their composition is treated like the composition with a 2-qubit gate.

DEFINITION 2.8 We will be doing plenty of work with spiders moving forward, so we will set some conventions: The **arity** of a spider is the number of inputs and outputs it has. If a spider has phase 0, we say it is **phase-less**, and often do not bother writing the 0. Namely, $\bigcirc := \textcircled{0}$ and $\bullet := \textcircled{0}$.

DEFINITION 2.9 The ZX Calculus admits a finite set of **rewrite rules**, or graphical equivalences for rewriting ZX diagrams. These are summarized in Table 1. For further explanation of these rules and some history on their names, please see [Wet20].

So far, it is not clear why we would choose to use ZX-Calculus over either the algebraic or circuit representation of a quantum circuit. This becomes clearer when we look at Table 1. Namely, Table 1 tells us that there is a short, finite set of rules for rewriting any ZX diagram to another. Paired with the fact that there are only two types (three if you count the H-box) of objects being manipulated, this notation allows us to reason about quantum computation at a higher level compared to the matrix or circuit representations. The following lemma is an example of this.

LEMMA 2.10 (COLOR CHANGE LEMMA) Any rewrite identity in ZX holds if we replace Z-spiders by X-spiders and vice versa.

¹Formally, the category of ZX diagrams, $\mathbf{ZX}$, is a compact closed symmetric monoidal category. The objects are non-negative integers and the morphisms ZX diagrams. The composition of morphisms correspond to the composition of ZX diagrams. The tensor product on objects satisfies $n \otimes m := n + m$ and the tensor product of two ZX diagrams is given by “stacking” the diagrams on top of another. Due to the existence of the cup and cap, this is a compact closed category. Interested readers should see [AC07] for more details.

Name	Rewrite rule	Description
Spider fusion (<i>f</i>)		Spiders of the same color connected by any number of wires fuse and their phases add.
Identity removal (<i>id</i>)		Spiders with one input, one output, and phase 0 are the identity matrix and can be removed.
Hadamard cancellation (<i>hh</i>)		Two connected Hadamards cancel each other.
π -commutation (π)		A π phase copies through a spider of the opposite color and flips its phase.
State copy (<i>cp</i>)		A π phase copies through a spider of the opposite color and flips its phase.
Color change (<i>cc</i>)		Commuting Hadamards through a spider changes the color.
Bi-algebra (<i>b</i>)		An adjacent phase 0 X and Z spider can commute past one another to create a complete bipartite graph with sides of the opposite color.
Hopf rule (<i>hf</i>)		If phase 0 spiders of opposite color are connected by multiple wires, we may remove the wires pairwise.

Table 1: ZX-Calculus Rewrite Rules

Proof. Consider a ZX diagram identity $D = D'$. For both: Apply (cc) on all spiders to swap all colors. All inputs and outputs connected to a spider gain a H-box, which are the same wires in D and D' . All wires between two spiders gain two H-boxes, which by (hh) cancel. Thus the color-swapped D is equivalent to the color-swapped D' . $\square$

THEOREM 2.11 Any ZX diagram is topologically invariant; only connectivity matters.

Proof. The proof surmounts to a tedious enumeration of all possible rearrangements of the same quantum circuit [CK17]. Like in quantum circuits, we often write ZX diagrams where inputs come from the left and outputs go to the right. This is simply convention. More formally, it suffices to maintain an ordered list of wires considered the input(s) and output(s). $\square$

EXAMPLE 2.12 The CNOT gate has the following ZX representation:

$$\begin{array}{c} \bullet \\ \hline \oplus \end{array} = \begin{array}{c} \circ \\ \hline \bullet \end{array}. \quad (2.9)$$

What Theorem 2.11 says is that

$$\left[\begin{array}{c} \circ \\ \diagdown \\ \bullet \end{array} \right] = \left[\begin{array}{c} \circ \\ \hline \bullet \end{array} \right] = \left[\begin{array}{c} \bullet \\ \diagup \\ \circ \end{array} \right]. \quad (2.10)$$

LEMMA 2.13 The ZX-Calculus is exactly universal for quantum computation², up to some complex scalar factor.

Proof. By Example 2.4 and Euler decomposition (Theorem A.12), we have that every single-qubit gate can be represented in the ZX-Calculus, up to some complex scalar factor. Example 2.12 gives us a representation of CNOT, a two-qubit entangling gate in the ZX-Calculus. By Theorem 1.3 of [BB01], arbitrary single-qubit gates with any entangling 2-qubit gate are exactly universal for quantum computation. $\square$

THEOREM 2.14 Any $c \in \mathbb{C}$ can be written as a ZX diagram.

Proof. Notice³

$$\left[\begin{array}{c} \pi \\ \hline \theta \end{array} \right] = e^{i\theta}, \quad \left[\begin{array}{c} \alpha \end{array} \right] = 1 + e^{i\alpha}, \quad \left[\begin{array}{c} \bullet \\ \hline \bullet \end{array} \right] = \frac{1}{2}, \quad \left[\begin{array}{c} \circ \end{array} \right] = 2. \quad (2.11)$$

Then

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2} (e^{-i\theta} (1 + e^{2i\theta})). \quad (2.12)$$

²See Definition 3.2 for a clarification of what exactly universal means.

³Here we use the fact that composition between ZX diagrams that are scalars is regular multiplication. This is a subtle implication of 2.5.

Thus we can write $\cos(\theta)$ as a ZX diagram. By the Archimedean property, for all $r \in \mathbb{R}$, there exists $k \in \mathbb{N}$ such that $2^k > |r|$. So we can write

$$r = 2^k \cos(\theta), \quad \theta = \arccos\left(\frac{r}{2^k}\right). \quad (2.13)$$

Then any real number can be written as a ZX diagram. Now, for any $c \in \mathbb{C}$, use the polar coordinates $c = \rho \cdot e^{i\theta}$ to obtain a ZX representation of c . Note that for $c = 0$, $\llbracket \pi \rrbracket = 1 - 1 = 0$. $\square$

THEOREM 2.15 The ZX-Calculus is exactly universal for quantum computation.

Proof. This follows by Theorem 2.14 and Lemma 2.13. $\square$

Furthermore, it is a well-known result [JPV20] that the rules in Table 1 are complete for quantum computation. That is, any identity that can be shown through the matrix representation can also be shown through ZX-Calculus. Equivalently, if two diagrams D, D' represent the same quantum computation, the diagrams can be transformed to one another using only the rules in Table 1. In fact, we may largely ignore scalar factors during this rewrite process. This is formalized below:

LEMMA 2.16 We may ignore scalar factors during ZX rewrites, provided that the desired scalar is restored at the end. This gives us the computation up to a global phase.

Proof. Given a quantum computation as a unitary U , we can obtain a diagram D where $\llbracket D \rrbracket = cU$, where $c \in \mathbb{C}$ is some non-zero scalar factor. We then have $\llbracket D \rrbracket \llbracket D \rrbracket^\dagger = c\bar{c}I$. Then

$$\frac{1}{|c|} \llbracket D \rrbracket = \frac{c}{|c|} U, \quad (2.14)$$

so we can recover U up to a global phase. $\square$

Computationally, it is often convenient to keep track of the product of all scalar factors in a diagram as a *number* rather than as a ZX diagram. We can use this extraneous data structure precisely because of Theorem 2.14, and as such we will recover the computation's global phase as well (unlike Lemma 2.16).

2.2 Clifford Circuits

DEFINITION 2.17 A **Clifford circuit** is a quantum circuit made of **Clifford gates**:⁴

$$\text{CNOT} := \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array}, \quad H := \text{---} \square \text{---} = \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} \text{---} \pi/2 \text{---} \\ | \\ \text{---} \pi/2 \text{---} \end{array}, \quad S := \text{---} \pi/2 \text{---}. \quad (2.15)$$

Furthermore, let H and S be known as the **local Clifford gates**.

⁴This is not the typical formal definition of Clifford circuits. However, we will soon show that this definition is equivalent to the usual one, i.e. as the circuits that normalize the Pauli group.

DEFINITION 2.18 A **Clifford ZX diagram** is a ZX diagram made up only of spiders whose phases are multiples of $\frac{\pi}{2}$. Furthermore, say a ZX diagram D is **unitary** if $\llbracket D \rrbracket = cU$ for some $c \in \mathbb{C}$ and unitary matrix U .

THEOREM 2.19 Clifford circuits are precisely the unitary Clifford ZX diagrams. That is, every Clifford circuit can be written as a unitary Clifford ZX diagram and vice versa.

Proof, part 1. It is clear that any Clifford circuit is a Clifford ZX diagram, since the diagrams of the Clifford gates only involve spiders with phases multiples of $\frac{\pi}{2}$. The reverse direction, however, will take more work. $\square$

To show the reverse direction, we construct an algorithm from Chapter 5 of [KW24] that allows us to convert any unitary Clifford ZX diagram into a Clifford circuit.

DEFINITION 2.20 A **Hadamard wire** is a wire with a single Hadamard, i.e. $\text{---}\square\text{---}$. To further disambiguate this from $\text{---}\bigcirc\text{---}$, we may write $\text{---}\square\text{---}$ in place of $\text{---}\square\text{---}$.

DEFINITION 2.21 A ZX diagram is **graph-like** when:

- (a) Every spider is a Z-spider.
- (b) Spiders are only connected via Hadamard edges.
- (c) There are no self loops or parallel / duplicate edges.
- (d) Every Z-spider is connected to at most one input and at most one output via a regular edge.
- (e) Every input and output is connected to a Z-spider.

THEOREM 2.22 Every ZX diagram can be efficiently rewritten as a graph-like diagram.

Proof. Let D be an arbitrary ZX diagram. Run the following algorithm:

1. Convert all X-spiders to Z-spiders via (cc) .
2. Cancel adjacent H with (hf) .
3. Fuse all connected spiders via (cc) until this cannot be done.
4. Remove parallel Hadamard edges via:

$$\begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(cc)}{=} \begin{array}{c} \text{---}\square\text{---} \end{array} \stackrel{(f)}{=} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(hf)}{=} \begin{array}{c} \text{---}\square\text{---} \end{array} \stackrel{(f)}{=} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(cc)}{=} \begin{array}{c} \text{---}\square\text{---} \end{array} \quad (2.16)$$

5. Remove self-loops via:

$$\begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(id)}{=} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(f)}{=} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array}, \quad (2.17)$$

$$\begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(f)}{\propto} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \stackrel{(hf)}{\propto} \begin{array}{c} \text{---}\bigcirc\text{---} \end{array} \quad (2.18)$$

6. Introduce Z-spiders and Hadamards using (id) and (hh) to ensure every input and output is connected to a Z-spider and no Z-spiders are connected to multiple inputs or multiple outputs.

After this, all spiders are Z-spiders. All edges between spiders are Hadamard edges, else the spiders would have been fused together. There is at most one edge between any two spiders, and there are no self-loops. Step 6 ensures (e) is satisfied. $\square$

Note that the above procedure applies to any ZX diagram, not just Clifford ones. As it turns out, when our starting ZX diagram is Clifford, we can simplify the diagram even further. To see how, we introduce the notion of a graph state and some operations on them.

DEFINITION 2.23 Say a spider in a ZX diagram is **internal** if it is not directly connected to an input or output wire, but it has a connection via other spiders to an input or output wire. Conversely, a spider is a **boundary** spider if it is directly connected to an input or output wire.

DEFINITION 2.24 For a simple, undirected graph $G = (V, E)$, the **graph state** $|G\rangle$ of G is defined

$$|G\rangle := \prod_{(v,w) \in E} CZ_{v,w} |+\rangle^{\otimes |V|}. \quad (2.19)$$

That is, prepare the $|+\rangle^{\otimes |V|}$ state and apply a CZ gate between every pair of adjacent vertices in G . If D is a ZX diagram and graph state, we write $G(D) = (V, E)$ to refer to the graph. Note that in the ZX Calculus,

$$|+\rangle \propto \text{---} \bigcirc \text{---}, \quad CZ = \begin{array}{c} \text{---} \bigcirc \text{---} \\ | \\ \text{---} \bigcirc \text{---} \end{array}. \quad (2.20)$$

EXAMPLE 2.25 The triangle graph has the following circuit and ZX graph state:

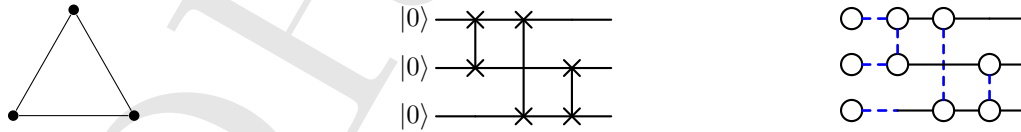


Figure 1: The triangle graph and its circuit and ZX graph state.

DEFINITION 2.26 A **graph state with local Cliffords (GSLC)** is a graph state to which some single-qubit Clifford unitaries have been applied on its outputs.

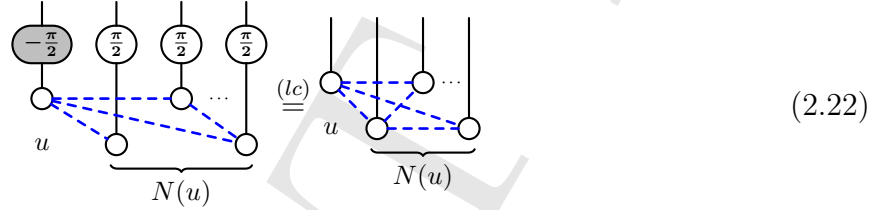
THEOREM 2.27 Every Clifford ZX diagram can be written in GSLC form.

To prove Theorem 2.27, we need to show that all internal spiders in a Clifford ZX diagram in graph-like form can be removed, perhaps by changing the phases of some of the boundary spiders and adding local Cliffords. To do this we need the following graph operations.

DEFINITION 2.28 Let G be a graph and $u \in V(G)$. The **local complementation** of G around u , written $G \star u$, is the graph obtained by complementing the subgraph induced by the neighbors of u . For distinct $v, w \in V(G)$, let $e_{v,w}$ be the edge between them. Then

$$e_{v,w} \in E(G \star u) \iff \begin{cases} e_{v,w} \notin E(G) & \text{if } v, w \in N_G(u), \\ e_{v,w} \in E(G) & \text{otherwise.} \end{cases} \quad (2.21)$$

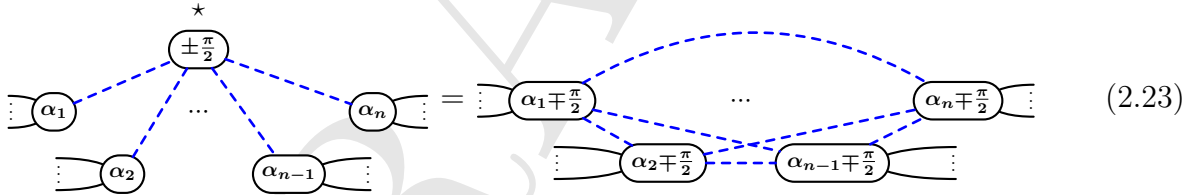
LEMMA 2.29 Let D be a graph-state ZX diagram, and u some vertex in $G(D)$. Then we can obtain a diagram D' with $G(D') = G(D) \star u$ by applying a $R_X(-\pi/2)$ gate on the spider corresponding to u and $R_Z(\pi/2)$ on all spiders in $N_{G(D)}(u)$. Call this rule (lc) :



Thus by applying local Cliffords we can local complement a graph state.

Proof. This was originally proved in Theorem 2 of [09]. The proof surmounts to a clever decomposition of the H-box as Z and X-spiders, which after some application of the rules in Table 1 gives the desired result. $\square$

LEMMA 2.30 The following holds:



Here, the $\star$ above $(\pm \frac{\pi}{2})$ indicates a local complementation at this vertex.

Proof. We provide a proof sketch: Let u denote the vertex we local complement around. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z-spiders. Apply local complementation in reverse around u . Then u will have $(\pm \frac{\pi}{2}) - (\pm \frac{\pi}{2}) = \bigcirc$ —dangling from it. Apply (cp) to push the $\bigcirc$ —through and disconnect u . This gives the RHS. For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

DEFINITION 2.31 Let G be a graph, $u, v \in V(G)$, and $e_{uv} \in E(G)$. The pivot of G along uv , denoted $G \wedge uv$, is defined as

$$G \wedge uv := G \star u \star v \star u \quad (2.24)$$

or equivalently

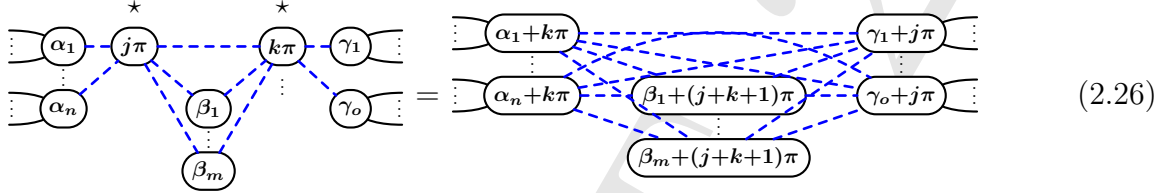
$$G \wedge uv = G \star v \star u \star v. \quad (2.25)$$

Proof. This is a standard proof by cases, seen for example in Definition 4.3 of [Dun+20]. $\square$

LEMMA 2.32 Given a graph G , we can pivot around uv for $u, v \in V(G)$ by applying Hadamard on u and v and Z on all vertices in $N_G(u) \cap N_G(v)$, inducing a swap gate between u and v .

Proof. The proof is surprisingly involved and can be found in Proposition 3.1 of [DP14]. $\square$

LEMMA 2.33 The following holds:

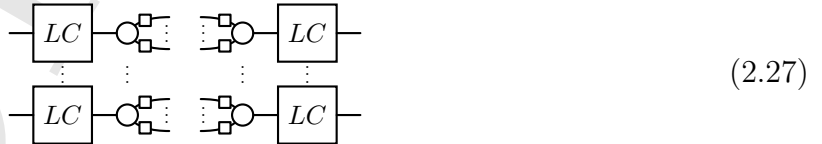


Note that the vertices $\beta_1, \dots, \beta_m$ may have other wires, but we suppress them here to reduce clutter.

Proof. We provide a proof sketch: Let u denote the vertex with phase $j\pi$ and v the vertex with phase $k\pi$. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z-spiders. Apply pivoting on uv . Then u will have $(k\pi)$ — $\square$ dangling from it (note the change from j to k due to the pivoting's swap gate). Apply (cc) and (cp) to obtain the RHS after fusing with (f) . For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

Proof of Theorem 2.27. First note that by Theorem 2.11, we can topologically morph the inputs of the Clifford circuit to face rightwards, like outputs, and that this has no effect on the operation represented by the diagram. Now, start by converting a Clifford ZX diagram to graph-like form. Apply Lemma 2.30 and 2.33 to remove all internal spiders from the diagram, perhaps appending local Clifford gates to the output wires. This gives a circuit in GSLC form. $\square$

LEMMA 2.34 The GSLC form of a CLifford ZX diagram may be written as follows:



Here $\boxed{LC}$ refers to circuits of the form $H^i S^j$ for some $i \in [0, 1]$ and $j \in [0, 1, 2, 3]$.

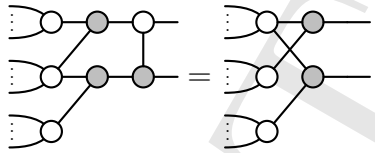
Proof. First note that we may topologically morph the GSLC state such that input wires face left and output wires face right. Now, note that the local Clifford generated must be of the form $H^i S^j$. Namely, when applying Lemma 2.30, we only modify the phase of boundary spiders. When we apply Lemma 2.33, we may add a Hadamard gate to an input or output wire. Since two Hadamards cancel, each input or output wire has at most one Hadamard. Since the phase of each boundary spider is a multiple of $\frac{\pi}{2}$ modulo 2π , unfusing the phase requires at most three S gates. $\square$

LEMMA 2.35 Let U be a Clifford unitary. Then U can be written as a circuit consisting of 8 layers made out of only the following gates in order:

$$H \quad S \quad CZ \quad CNOT \quad H \quad CZ \quad S \quad H \quad (2.28)$$

Proof. This proof is taken from Theorem 5.3.11 of [KW24]. We provide a proof sketch: Start from Lemma 2.34. This explains the outer H and S layers. Now, apply (f) in reverse to obtain a CZ , representing an edge between two vertices both on the left or both on the right. This explains the two CZ layers. Then, apply (cc) to obtain the H layer in between $CNOT$ and CZ .

What remains is a bipartite graph whose left vertices are phase-less Z -spiders and right vertices phase-less X -spiders. As it turns out, because our diagram is unitary, this bipartite graph can be decomposed into a circuit made only of $CNOT$ gates via the following identity:

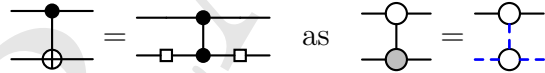


$$(2.29)$$

The proof of this surmounts to book-keeping and a clever use of (b) . Please see Lemma 4.2.11 of [KW24] for a step-by-step proof. $\square$

Proof of Theorem 2.27 and Proof of Theorem 2.19, part 2. Given a unitary Clifford ZX diagram, do the following:

1. Convert it to graph-like form, using Theorem 2.22.
2. Use Lemma 2.30 and Lemma 2.33 to remove internal spiders and bring the diagram to GSLC form.
3. Use Lemma 2.35 to obtain a Clifford circuit for the diagram, by noting that



$$(2.30)$$

Since we have provided an injective map in both directions, the proof of Theorem 2.19 is complete. $\square$

COROLLARY 2.36 Any n -qubit Clifford unitary can be represented by an equivalent circuit in at most $4n(n + 3/2) - 1$ Hadamard, S , and $CNOT$ gates.

Proof. Look at Lemma 2.35. Each Hadamard layer has at most n gates, and there are three of those. Each S gate layer has at most $3n$ gates, and there are two of them. A circuit of CZ gates can have at most one CZ gate for every pair of qubits, as CZ gates commute past each other. Thus the CZ layer has at most $n(n - 1)/2$ gates. CZ s are made up of one $CNOT$ and two Hadamards, so this contributes $3 \cdot n(n - 1)/2$ gates. Finally, the $CNOT$ circuit is essentially the number of distinct invertible maps from $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. It is left as an exercise to the reader to see that there are at most $n^2 - 1$ such maps. Thus, in total at most

$$9n + 2 \cdot 3 \cdot \frac{n(n - 1)}{2} + n^2 - 1 = 4n(n + 3/2) - 1 \quad (2.31)$$

gates are used. $\square$

THEOREM 2.37 Let U be a q -qubit Clifford unitary with N gates. We can rewrite this as a ZX diagram in GSLC form in $O(Nq^2)$ time, if $N \geq q$. Further, we can calculate any amplitude

$$\langle x_1, \dots, x_n | U | 0, \dots, 0 \rangle \quad (2.32)$$

in $O(Nq^2)$ time. This is practically equivalent to the **Gottesman-Knill theorem**. If U is already available in GSLC form, this can be done in $O(q^3)$ time.

Proof. The first comes from analyzing the runtime of the algorithm in Theorem 2.19's part 2 proof. The second comes from the fact that, a post-selection of outcomes $|x_1, \dots, x_n\rangle$ is equivalent to “plugging” the output wires of a GSLC diagram with the corresponding spiders. The third claim follows from a similar observation. Please see Proposition 5.4.4 of [KW24] for a more detailed proof. $\square$

2.3 The Stabilizer Formalism

When we view the Clifford circuits of the previous section more algebraically, we find interesting connections to the following objects:

DEFINITION 2.38 The **Pauli matrices** or **Pauli gates** refer to the following gates:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (2.33)$$

DEFINITION 2.39 Say a matrix A is a **stabilizer** of the state $|\psi\rangle$ when $|\psi\rangle$ is a $+1$ eigenvector of A , i.e. $A|\psi\rangle = |\psi\rangle$.

DEFINITION 2.40 The n -qubit Pauli group is defined as follows:

$$\mathcal{P}_n := \{i^k P_1 \otimes \dots \otimes P_n \mid k \in \{0, 1, 2, 3\}, P_j \in \{I, X, Y, Z\}\}.$$

Call the elements of $\mathcal{P}_n$ as **Pauli strings**, written as vectors of length n :

$$\vec{P} := i^k P_1 \otimes \dots \otimes P_n.$$

Proof. We need to show that this is indeed a group. Associativity follows from matrix multiplication. I is the identity. Each Pauli matrix is its own inverse, i.e. $X^2 = Y^2 = Z^2 = 1$, and it is easy to see that the inverses of $i^k P$ exist for all k and P . Finally, since two Pauli matrices $P_1 \neq P_2$ anti-commute, i.e. $P_1 P_2 = -P_2 P_1$, an exhaustive multiplication table shows the group is closed and finite. $\square$

THEOREM 2.41 For any Clifford unitary U and Pauli string $\vec{P}$ of compatible dimensions, there exists a pauli string $\vec{Q}$ such that $U \vec{P} U^\dagger = \vec{Q}$.

Proof. It suffices to check that, when we commute a basis of the Pauli strings through the Hadamard, S, and CNOT gate, we will end up with another Pauli string. Notice that $\{X, Z\}$ is a basis for $\mathcal{P}_1$, up to scalar factors. Now note that

$$\text{---} \circ \pi \text{---} \square \text{---} \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \square \text{---} \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}. \quad (2.34)$$

For the S gate,

$$\text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{(f)}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---} \circ \pi \text{---}. \quad (2.35)$$

For the CNOT gate,

$$\begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad (2.36)$$

$$\begin{array}{ccc} \text{---} \circ \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \pi \text{---} \circ \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \pi \text{---} & & \text{---} \circ \pi \text{---} \end{array}. \quad \square$$

DEFINITION 2.42 For a group $\mathcal{S} \subseteq \mathcal{P}_n$, define the **stabilizer subspace**⁵ of $\mathcal{S}$ as

$$\text{Stab}(\mathcal{S}) := \{|\psi\rangle \mid \vec{P}|\psi\rangle = |\psi\rangle, \forall \vec{P} \in \mathcal{S}\}. \quad (2.37)$$

If $\text{Stab}(\mathcal{S}) \neq \{0\}$, call $\mathcal{S}$ a **stabilizer group**.

Proof. If A, B are stabilizers of $|\psi\rangle$, then $AB|\psi\rangle = A|\psi\rangle = |\psi\rangle$ so AB is also a stabilizer. Furthermore,

$$A|\psi\rangle = |\psi\rangle \implies |\psi\rangle = A^{-1}|\psi\rangle$$

so A^{-1} is also a stabilizer of $|\psi\rangle$. Thus $\mathcal{S}$ is a subgroup. On the other hand, if two states $|\psi\rangle$ and $|\phi\rangle$ are stabilized by A , then

$$A(\lambda|\psi\rangle + \mu|\phi\rangle) = \lambda A|\psi\rangle + \mu A|\phi\rangle = \lambda|\psi\rangle + \mu|\phi\rangle, \quad (2.38)$$

for any $\lambda, \mu \in \mathbb{C}$. Since 0 is trivially in every $\text{Stab}(\mathcal{S})$, this is indeed a subspace. $\square$

LEMMA 2.43 If a subgroup $\mathcal{S} \subseteq \mathbb{P}_n$ is a stabilizer subgroup, then:

1. $-I \notin \mathcal{S}$,
2. $\vec{P}^2 = I$ for all $\vec{P} \in \mathcal{S}$, and
3. $\mathcal{S}$ is commutative.

⁵This is a strictly more restrictive definition of stabilizer subspaces and stabilizer groups than the definitions found in algebra, for instance. However, since we are only dealing with pure states, and are only concerned with the stabilizer subspaces of $\mathcal{P}_n$, this definition suffices.

Proof. If $-I \in \mathcal{S}$, then all $|\psi\rangle \in \text{Stab}(\mathcal{S})$ has $|\psi\rangle = -I|\psi\rangle = -|\psi\rangle$ such that $\text{Stab}(\mathcal{S}) = \{0\}$, a contradiction. An exhaustive multiplication table for $\mathcal{P}_1$ reveals that if $\vec{P}^2 \neq I$, then $\vec{P}^2 = -I$ so again $\text{Stab}(\mathcal{S}) = \{0\}$. Thus, each $P \in \mathcal{S}$ must have $P^2 = Q^2 = I$. Then $P^\dagger = P$, since $P^{-1} = P$ and its phase cannot be $\pm i$. Then for any $P, Q \in \mathcal{S}$, $PQ = (PQ)^\dagger = Q^\dagger P^\dagger = QP$. $\square$

THEOREM 2.44 This is Theorem 1 of [AG04]. Given an n -qubit state $|\psi\rangle$, the following are equivalent:

1. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, and S gates only.
2. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, S, and measurement gates only.
3. $|\psi\rangle$ is stabilized by exactly 2^n Pauli operators.
4. $|\psi\rangle$ is uniquely determined by $\text{Stab}(|\psi\rangle)$.

Thus, we can call any circuit consisting entirely of Clifford gates and measurements a **stabilizer circuit**, and any state obtainable by applying a stabilizer circuit to $|0\rangle^{\otimes n}$ a **stabilizer state**. Equivalently, a stabilizer state is a state $|\psi\rangle \in \mathbb{C}^{2^n}$ such that $\vec{P}|\psi\rangle = |\psi\rangle$ for all $\vec{P} \in \mathcal{S} \subseteq \mathcal{P}_n$.

Proof. A concise proof is available in [AG04], which itself relies on the Gottesman-Knill theorem (Theorem 2.37), whose ZX-proof can be found in Chapter 5 of [KW24] and non-ZX-proof in the original paper [Got98]. $\square$

LEMMA 2.45 Note that $|\mathcal{P}_n| = 4^{n+1}$. Furthermore, the number of stabilizer states on n qubits is:

$$2^n \prod_{i=0}^{n-1} (2^{n-i} + 1). \quad (2.39)$$

Proof. The first claim follows almost immediately from Definition 2.40. By Theorem 2.44, to obtain the second claim we can equivalently count the number of ordered generating sets for stabilizer groups, divided by the number of generating sets for a given stabilizer. This is the proof used in Proposition 2 of [AG04]. $\square$

Informally, what this stabilizer formalism tells us is that states obtainable by applying Clifford circuits to $|0\rangle^{\otimes n}$ are precisely the pure stabilizer states. This is significant for a whole host of reasons, not the least of which is the tableau formalism below. This tableau formalism essentially tells us that we can keep track of a Clifford computation by keeping track of changes to the stabilizer of the state.

DEFINITION 2.46 The tableau of a state $|\psi\rangle$ is a matrix consisting of binary variables x_{ij} and z_{ij} for $i \in [2n]$ and $j \in [n]$, and r_i for all $i \in [2n]$:

$$\left[\begin{array}{ccc|ccc|c} x_{11} & \cdots & x_{1n} & z_{11} & \cdots & z_{1n} & r_1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & \cdots & x_{nn} & z_{n1} & \cdots & z_{nn} & r_n \\ \hline x_{(n+1)1} & \cdots & x_{(n+1)n} & z_{(n+1)1} & \cdots & z_{(n+1)n} & r_{n+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{(2n)1} & \cdots & x_{(2n)n} & z_{(2n)1} & \cdots & z_{(2n)n} & r_{2n} \end{array} \right] \quad (2.40)$$

where

$$x_{ij} = \begin{cases} 1 & \text{if } P_{ij} = X \text{ or } P_{ij} = Y, \\ 0 & \text{otherwise,} \end{cases} \quad (2.41)$$

$$z_{ij} = \begin{cases} 1 & \text{if } P_{ij} = Y \text{ or } P_{ij} = Z, \\ 0 & \text{otherwise,} \end{cases} \quad (2.42)$$

$$r_i = \begin{cases} 1 & \text{if } R_i \text{ has negative phase,} \\ 0 & \text{otherwise.} \end{cases} \quad (2.43)$$

THEOREM 2.47 Using the tableau representation, we can efficiently simulate CNOT, H, and S gates. This is an alternative method of simulation compared to Theorem 2.37.⁶

Proof. The proof surmounts to a clever bookkeeping of the stabilizers of a state as Clifford gates are applied to it. This proof can be found in [AG04]. Here we provide some intuition for why this is the case. The key observation is that each Pauli gate and I can be represented by two classical bits (x, z) :

$$\begin{array}{c|cc} P & x & z \\ \hline I & 0 & 0 \\ X & 1 & 0 \\ Z & 0 & 1 \\ Y & 1 & 1 \end{array} \quad (2.44)$$

Note that we represent Y as $(1, 1)$ even though $Y = iXZ$. This is due to Lemma 2.43; each Pauli stabilizer cannot have a phase of $\pm i$, but may have a sign ± 1 . Thus, for a Pauli string $\vec{P}$ of length n , we can capture its data using two bits for each of the n qubits, plus an extra bit r for the overall sign. The tableau has $2n$ rows for bookkeeping and algorithmic reasons.⁷ $\square$

This algorithm may be improved further. See [SSY25] for such an example.

⁶As it turns out, both methods of simulation, i.e. the ZX-Calculus GSLC algorithm and the tableau representation, will be used in this thesis.

⁷The first n rows keep track of the *de-stabilizers* of the state, while the next n rows keep track of the stabilizers. See [AG04] for more details.

3 Measurement-Based Quantum Computation

3.1 Universality

In this thesis, we are primarily concerned with the universality of the comparability grid for measurement-based quantum computation. It follows that we need a precise definition of universality to formally state our problem.

DEFINITION 3.1 For every $n \in \mathbb{N}$, Let $SU(n)$ denote the special unitary group under matrix multiplication consisting of $n \times n$ unitary matrices with determinant 1. It is a normal subgroup of the unitary group $U(n)$, consisting of $n \times n$ unitary matrices.

DEFINITION 3.2 A set of gates $\mathcal{S}$ is (efficiently) **universal** if any $U \in SU(2^n)$, for any $n \in \mathbb{N}$, that can be generated (in $\text{poly}(n)$ gates) by an arbitrary gate set $\mathcal{S}'$, can also be generated by the gate set $\mathcal{S}$ (in $\text{poly}(n)$ gates) [Nes+07]. We say a set of gates $\mathcal{S}$ is d -qubit universal if the above holds for $U \in SU(2^d)$.

In this thesis we will only focus on universality, not efficient universality. Thus, it may take exponential or more gates to simulate certain unitaries U . Also, in practice, the application of U up to some measure and error ϵ suffices. This is called **approximate universality**. This thesis will focus solely on **exact universality**, where we desire no error, rather than approximate universality. Henceforth, all references to universality will refer to exact, and not necessarily efficient, universality.

Furthermore, note that definition 3.2 is only concerned with implementing arbitrary unitaries, not necessarily arbitrary computation. That is, it does not specify whether the inputs and outputs are classical or quantum in nature. This matters because $|0\rangle$ and $-|0\rangle$ have the same measurement distribution and classical output, but $|0\rangle \neq -|0\rangle$. This difference is called **global phase**, i.e. a scalar $e^{i\theta}$ multiplied to an entire state.

DEFINITION 3.3 We use the following types of **universal computation** from [Nes+07]:

- **CC-universal**: For every $U \in SU(n)$, the model can reproduce the distribution of local measurements performed on $U|0^n\rangle$.
- **QC-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can reproduce the distribution of local measurements performed on $U|\phi\rangle$.
- **CQ-universal**: For every $U \in SU(2^n)$, the model can produce the quantum state $|\phi\rangle = U|0^n\rangle$.
- **QQ-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can produce the quantum state $U|\phi\rangle$.

These distinctions allow us to more finely categorize universality. For example, we may restate Lemma 2.16 as follows: Ignoring scalar factors during ZX rewrites, provided that the desired scalar is restored at the end, makes ZX-Calculus at most QC-universal, while keeping track of scalars throughout gives QQ-universality.

The first three CNOTs are equivalent to a swap gate. Thus

$$\begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \\ \bullet \\ \oplus \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \\ \oplus \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ H|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} \quad (3.4)$$

The final state before measurement is

$$\text{CNOT}(|+\rangle \otimes H|\phi\rangle) = \frac{1}{\sqrt{2}} (|0\rangle \otimes H|\phi\rangle + |1\rangle \otimes XH|\phi\rangle). \quad (3.5)$$

In other words, a measurement of the first qubit in the **computational basis** ($\{|0\rangle, |1\rangle\}$) will tell us whether we have a $H|\phi\rangle$ or $XH|\phi\rangle$ on the output wire.

Now, suppose our intention in the above circuit was to apply a Hadamard gate to $|\phi\rangle$. Then there is a 0.5 probability that we apply XH instead of H , which is indicated when we measure a 1. How can we get rid of this additional X ?

EXAMPLE 3.6 Define the **teleportation gadget** as the following circuit.

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{R_Z(\theta)} \\ \boxed{H} \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ HR_Z(\theta)|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array}. \quad (3.6)$$

Let $m \in \{0, 1\}$ be the measurement outcome of measuring in the computational basis. Then we apply $X^m HR_Z(\theta)$ to $|\phi\rangle$.

Proof. We show that the RHS circuit is indeed the LHS circuit. First note that, by (f), the $R_Z(\theta)$ gate commutes through the CZ unchanged. The rest of the analysis is identical to Example ??.

EXAMPLE 3.7 Now, suppose we chain together two teleportation gadgets, taking the output of one as the input ϕ to another. Then we are applying the following gate:

$$X^{m_2} HR_Z(\theta_2) X^{m_1} HR_Z(\theta_1) \stackrel{(\pi)}{=} X^{m_2} H X^{m_1} R_Z((-1)^{m_1} \theta_2) HR_Z(\theta_1) \quad (3.7)$$

$$\stackrel{(cc)}{=} X^{m_2} Z^{m_1} HR_Z((-1)^{m_1} \theta_2) HR_Z(\theta_1) \quad (3.8)$$

Chaining three gadgets, we are applying the following gate:

$$X^{m_3} HR_Z(\theta_3) X^{m_2} HR_Z(\theta_2) X^{m_1} HR_Z(\theta_1) \quad (3.9)$$

$$= X^{m_3} HR_Z(\theta_3) (X^{m_2} Z^{m_1} HR_Z((-1)^{m_1} \theta_2) X^{m_1} HR_Z(\theta_1)) \quad (3.10)$$

$$\stackrel{(\pi)}{\stackrel{(cc)}}{=} X^{m_3+m_1} Z^{m_2} (HR_Z((-1)^{m_2} \theta_3) HR_Z((-1)^{m_1} \theta_2) X^{m_1} HR_Z(\theta_1)). \quad (3.11)$$

Since $X^2 = Z^2 = I$, it is not too hard to see that chaining this gadget any number of times will result in an overall Pauli error of $X^m Z^n$ for some bits m, n .

The formal definition of the MBQC protocol incorporates these errors into the definition.

DEFINITION 3.8 We use the formal definition from [Bro+07], known as the **Measurement Calculus**. Denote an arbitrary single qubit state by

$$|+\theta,\phi\rangle := \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.12)$$

and denote its orthogonal state by

$$|+\theta,\phi\rangle := \cos\left(\frac{\theta}{2}\right)|0\rangle - e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.13)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. A computation in this model is a series of the following deterministic **commands**:

1. Preparation: 1-qubit preparations N_i , to prepare qubit i in the state $|+\rangle$;
2. Entanglement: 2-qubit entangling operators $E_{i,j} := Z_{i,j}$;
3. Measurement: 1-qubit measurements $M_i^{\lambda,\alpha}$ on planes $\lambda \in \{(X,Y), (X,Z), (Y,Z)\}$, defined as projections into

- (i) $|\pm_{(X,Y),\alpha}\rangle := |+\pi/2,\alpha\rangle$ for $\lambda = (X,Y)$,
- (ii) $|\pm_{(X,Z),\alpha}\rangle := |+\alpha,0\rangle$ for $\lambda = (X,Z)$, and
- (iii) $|\pm_{(Y,Z),\alpha}\rangle := |+\alpha,\pi/2\rangle$ for $\lambda = (Y,Z)$,

with the convention that $|+\theta,\phi\rangle\langle+\theta,\phi|$ corresponds to the outcome 0 and $|-\theta,\phi\rangle\langle-\theta,\phi|$ corresponds to 1; and

4. Correction: 1-qubit Pauli corrections X_i and Z_i .

The measurements $M_i^{\lambda,\alpha}$ may depend on **signals**, namely sums of some outcomes of previous measurements mod 2. These represent the Pauli errors of the form $X^m Z^n$ being carried through, for some bits m, n . A computation can then be expressed as a series of preparation, entanglement, measurement, and correction commands, strictly in this order, satisfying the following conditions:

- (a) No command depends on an outcome not yet measured;
- (b) No command acts on a qubit already measured;
- (c) No command acts on a qubit not yet prepared, unless it is an input qubit; and
- (d) A qubit i is measured if and only if i is not an output.

Equivalently, a computation may be defined as a **measurement pattern** on a some graph state $|G\rangle$. A measurement pattern $\mathcal{B} = (V, I, O, S)$ consists of

- a finite set of qubits $V \subset V(G)$,

- an input set $I \subset V(G)$, possibly overlapping with
- an output set $O \subset V(G)$,
- and a finite sequence of measurement and correction commands S .

that satisfy the above conditions (a) to (d).

Note that the measurement bases available in the Measurement Calculus model are different from the measurements in Examples 3.5 and 3.6. Thus, it will be convenient to find the operations applied by measurements in the XY, YZ, and XZ planes.

LEMMA 3.9 Let $|\psi\rangle = a|0\rangle + b|1\rangle$ be an arbitrary one-qubit input state. Prepare a second qubit $|+\rangle$, apply CZ, and measure the first qubit in a state

$$|m\rangle = u|0\rangle + v|1\rangle. \quad (3.14)$$

Conditioned on this measurement outcome, the unnormalized output state on the second qubit is

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle. \quad (3.15)$$

In particular, for the standard measurement planes we obtain:

Plane	Measurement state	ZX effect	Unnormalized output	Unitary?
XY	$\frac{1}{\sqrt{2}}(0\rangle + e^{-i\alpha} 1\rangle)$		$\frac{1}{\sqrt{2}}HP(\alpha) \psi\rangle$	Always
YZ	$c 0\rangle + is 1\rangle$		$H \begin{pmatrix} c & 0 \\ 0 & -is \end{pmatrix} \psi\rangle$	Only if $ c = s $
XZ	$c 0\rangle + s 1\rangle$		$H \begin{pmatrix} c & 0 \\ 0 & s \end{pmatrix} \psi\rangle$	Only if $ c = s $

where $P(\alpha) = \text{Diag}(1, e^{i\alpha}) = \llbracket \textcircled{\alpha} \rrbracket$, $c = \cos(\frac{\alpha}{2})$, and $s = \sin(\frac{\alpha}{2})$.

Proof. Write $|\psi\rangle = a|0\rangle + b|1\rangle$. Prepare a second qubit $|+\rangle$ and apply CZ:

$$CZ(|\psi\rangle|+\rangle) = CZ((a|0\rangle + b|1\rangle)|+\rangle) = a|0\rangle|+\rangle + b|1\rangle|-\rangle. \quad (3.16)$$

Now project the first qubit onto $|m\rangle = u|0\rangle + v|1\rangle$. The resulting unnormalized state is

$$(\langle m| \otimes I)CZ(|\psi\rangle|+\rangle) = (\langle m| \otimes I)(a|0\rangle|+\rangle + b|1\rangle|-\rangle) \quad (3.17)$$

$$= a\langle m|0\rangle|+\rangle + b\langle m|1\rangle|-\rangle \quad (3.18)$$

$$= a\bar{u}|+\rangle + b\bar{v}|-\rangle \quad (3.19)$$

$$= H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle. \quad (3.20)$$

For the XY -plane state, $|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{-i\alpha}|1\rangle)$,. So $\bar{u} = \frac{1}{\sqrt{2}}$, and $\bar{v} = \frac{e^{i\alpha}}{\sqrt{2}}$.. Hence

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & e^{i\alpha}/\sqrt{2} \end{pmatrix} |\psi\rangle = \frac{1}{\sqrt{2}} H \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix} |\psi\rangle = \frac{1}{\sqrt{2}} HP(\alpha) |\psi\rangle. \quad (3.21)$$

For the YZ -plane state, $|m\rangle = c|0\rangle + is|1\rangle$,. Then $\bar{u} = c$, and $\bar{v} = -is$.. Therefore

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} c & 0 \\ 0 & -is \end{pmatrix} |\psi\rangle. \quad (3.22)$$

For the XZ -plane state, $|m\rangle = c|0\rangle + s|1\rangle$,. Hence $\bar{u} = c$, and $\bar{v} = s$. This gives

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} c & 0 \\ 0 & s \end{pmatrix} |\psi\rangle. \quad (3.23)$$

Finally, since H is unitary, $H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix}$ is proportional to a unitary iff $\begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix}$ is proportional to a unitary. A diagonal matrix $D = \begin{pmatrix} r_0 & 0 \\ 0 & r_1 \end{pmatrix}$ is proportional to a unitary iff

$$D^\dagger D = \begin{pmatrix} |r_0|^2 & 0 \\ 0 & |r_1|^2 \end{pmatrix} = \lambda I \quad (3.24)$$

for some $\lambda > 0$, which holds exactly when $|r_0| = |r_1|$. Thus the XY case is always unitary, while the YZ and XZ cases are unitary only when $|c| = |s|$. $\square$

LEMMA 3.10 The Pauli eigenstates occur as special cases of XY , XZ , and YZ plane measurements:

Pauli effect	Two-plane form	State	ZX effect
$\langle +_X $	$\langle +_{XY,0} $	$\langle + = \frac{1}{\sqrt{2}}(\langle 0 + \langle 1)$	$\bigcirc -$
$\langle -_X $	$\langle +_{XY,\pi} $	$\langle - = \frac{1}{\sqrt{2}}(\langle 0 - \langle 1)$	$\bigcirc(\pi) -$
$\langle +_Y $	$\langle +_{XY,\pi/2} $	$\langle -i := \frac{1}{\sqrt{2}}(\langle 0 - i \langle 1)$	$\bigcirc(-\frac{\pi}{2}) -$
$\langle -_Y $	$\langle +_{XY,-\pi/2} $	$\langle i := \frac{1}{\sqrt{2}}(\langle 0 + i \langle 1)$	$\bigcirc(\frac{\pi}{2}) -$
$\langle +_Z $	$\langle +_{XZ,0} = \langle +_{YZ,0} $	$\langle 0 $	$\bullet -$
$\langle -_Z $	$\langle +_{XZ,\pi} = \langle +_{YZ,\pi} $	$\langle 1 $	$\bullet(\pi) -$

Consequently, X - and Y -basis measurements are contained in the XY -plane model, while Z -basis measurements require the XZ (or YZ) plane,

Proof. The XY -plane cases follow directly from substituting the relevant angles:

$$|+_{XY,0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i0} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+_X\rangle, \quad (3.25)$$

$$|+_{XY,\pi}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-_X\rangle. \quad (3.26)$$

Similarly,

$$|+_{XY,\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) = |+_Y\rangle, \quad (3.27)$$

$$|+_{XY,-\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{-i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - i |1\rangle) = |-_Y\rangle. \quad (3.28)$$

For the Z -basis states, use the XZ plane:

$$|+_{XZ,0}\rangle = \cos(0) |0\rangle + \sin(0) |1\rangle = |0\rangle = |+_Z\rangle, \quad (3.29)$$

$$|+_{XZ,\pi}\rangle = \cos \frac{\pi}{2} |0\rangle + \sin \frac{\pi}{2} |1\rangle = |1\rangle = |-_Z\rangle. \quad (3.30)$$

Finally, every XY -plane state has the form

$$|+_{XY,\alpha}\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{e^{i\alpha}}{\sqrt{2}} |1\rangle. \quad (3.31)$$

Hence both computational-basis amplitudes have magnitude $1/\sqrt{2}$. Therefore no XY -plane state can equal $|0\rangle$ or $|1\rangle$, even up to global phase. Thus Z -basis measurements are not contained in the XY -plane measurement model. $\square$

In this thesis, we are interested in whether there exist measurement patterns on comparability grids that result in universal quantum computation. It remains to formally define what it means to use comparability grids as a resource for universal quantum computation.

3.3 Resource States

DEFINITION 3.11 A **resource state** is a state, typically containing entanglement, on which a measurement pattern can be applied to conduct measurement-based quantum computation. Under the Measurement Calculus (Definition 3.8), a resource state is a graph state.

DEFINITION 3.12 A **family** of resource states is some collection of resource states. A family of resource states Ψ is **universal for MBQC** if, for every n -qubit state $|\phi\rangle$, there exists some resource state $|\Psi_n\rangle \in \Psi$ and a measurement pattern $\mathcal{B}$ such that the output state is $|\phi\rangle$. We say a family of resource states is **n -qubit universal for MBQC** if this holds for all m -qubit $|\phi\rangle$ with $m \leq n$. When the meaning is clear, we may drop the phrase “for MBQC” and simply say a family of resource states is universal.

We start by reciting some well-known results related to graph states.

FACT 3.13 There is a unique quantum circuit construction for every graph state, up to reordering of C-Z.

Proof. By (f), CZ gates commute past each other. Then each pair of qubits (vertices) has zero or one CZs (edges) between them. $\square$

FACT 3.14 Let G be a graph with maximum degree Δ . Then $|G\rangle$ can be constructed in depth $\Delta + 1$.

Proof. Vizing's theorem states that any graph with maximum degree Δ admits an edge coloring with $\Delta + 1$ colors such that no two edges that share a vertex have the same color. All edges of the same color can be implemented in the graph state in the same layer of CZs. Thus, at most $\Delta + 1$ layers are needed. $\square$

FACT 3.15 Given a graph state $|G\rangle$, a measurement in the YZ plane is equivalent to vertex deletion.

Proof. Consider a graph state $|G\rangle$ in its ZX-Calculus representation.

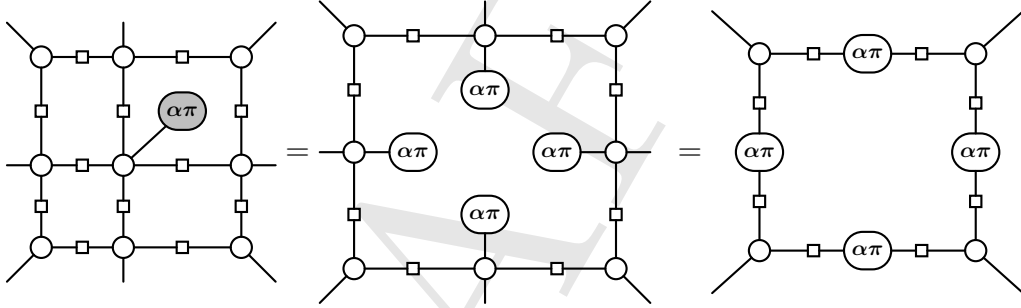


Figure 2: YZ plane measurements correspond to vertex deletion.

By Lemma 3.9, the ZX effect of a YZ plane measurement is a $\textcircled{a\pi}$ where a is the measurement outcome bit. By (cp), we may push the $\textcircled{a\pi}$ through the vertex being measured, deleting it in the process. The result is a graph state with the measured vertex removed, and an additional phase of $a\pi$ on all vertices connected to the measured vertex. $\square$

FACT 3.16 Given a graph state $|G\rangle$, applying $\sqrt{X} = -\textcircled{\pi/2}$ on a vertex v and $\sqrt{Z} = -\textcircled{\pi/2}$ on $N_G(v)$ is equivalent to a local complementation around v , and yields $|G \star v\rangle$. Applying a Y plane measurement on v instead gives $|G \star v - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.29 and the second is Lemma 2.30. $\square$

FACT 3.17 Given a graph state $|G\rangle$, applying H on vertices u, v and Z on $N_G(u) \cap N_G(v)$ gives $|G \wedge uv\rangle$. Applying a X plane measurement on both u and v instead gives $|G \wedge uv - u - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.32 and the second is Lemma 2.33. $\square$

DEFINITION 3.18 A $m \times n$ grid graph $G = (V, E)$ has

$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (3.32)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, |i_2 - i_1| + |j_2 - j_1| = 1\} \quad (3.33)$$

In other words, m rows and n columns with edges between nodes of taxicab distance 1. Denote such graphs as $\text{Grid}_{m \times n}$. Let Grid denote the family of all such graphs.

THEOREM 3.19 The Grid is universal.

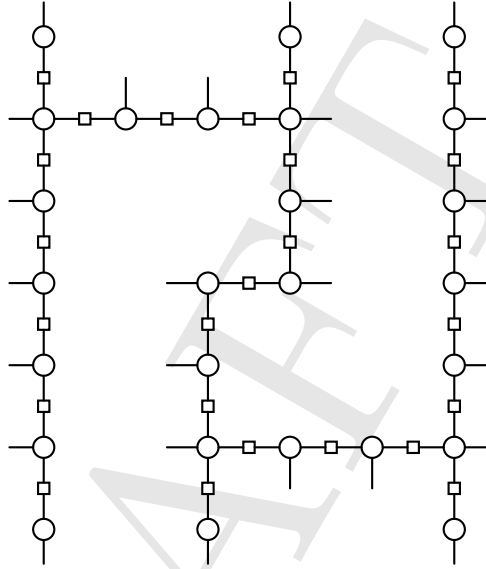


Figure 3: A Grid graph state with some vertices removed, creating three vertical “wires” for a three qubit computation in the MBQC model. The horizontal connections between wires may entangle the qubits, given the right measurements.

Proof. The full proof is quite involved and very tedious. The original proof is given in [RB01], and a more accessible proof is available in [RBB03]. A modern proof through the ZX-Calculus is available in Chapter 9 of [KW24].

There are three great difficulties in the proof. We informally discuss them. First, we need to be able to implement arbitrary single qubit gates. This is done by deleting (via Z plane measurements) vertices to create line graphs, each resulting in a chain like Example 3.6. It can be shown that this allows the application of arbitrary single qubit gates up to some Pauli error.

Second, we need to be able to entangle qubits. If the two line graphs containing the qubits we wish to entangle are adjacent, i.e. there are no other line graphs between them, then creating a orthogonal line of vertices between the wires may entangle them. Now, if they are not adjacent, we can swap qubits between lines (by implementing CNOT thrice) until they become adjacent.

Third, we need to ensure measurement is done in the correct order. For example, suppose we delete all vertices connecting the input and output vertices. Then this computation is deterministic and not invertible, and thus does not implement a unitary. As it turns out, there is a criterion called *flow* that ensures this does not happen, as well as variants of flow such as *gflow* and *Pauli flow* that further ensure that we can deal with the Pauli errors. Interested readers should see [MB23; Bac+21; KW24] for more details. $\square$

3.4 Universality Criteria

Given the difficulty of the proof of Theorem 3.19, we now turn our attention to the development of other techniques to determine the universality of a family of resource states. We start by recalling some useful facts regarding regular quantum circuit universality.

FACT 3.20 Any 1-qubit universal gate set and a 2 qubit entangling gate is universal [BB01].

Proof. The proof in [BB01] turns out to be quite technical, involving many tricks from Lie Algebra. It surmounts to proving the following restatement of the theorem: Let V be a 2-qubit gate. The following are equivalent:

1. The collection of all 1-qubit gates $\mathcal{A}$ together with V is QQ-universal.
2. The collection of all 1-qubit gates $\mathcal{A}$ together with V is exactly QQ-universal.
3. V is not primitive, or **imprimitive**.

Here, say V is **primitive** iff $V = S \otimes T$ or $V = (S \otimes T)P$ for some 1-qubit gates S and T , with P being the swap gate. $\square$

THEOREM 3.21 Any universal gate set must include at least one multi-qubit entangling gate, as it is impossible to entangle using only single-qubit gates.

Proof. This follows from Fact 3.20. $\square$

THEOREM 3.22 Let **Line** denote the family of graphs made out of line graphs. Then **Line** is not a universal resource.

Proof. This follows from Theorem 3.21. $\square$

We now turn to universality criteria specifically for MBQC. We start by addressing why we only consider CQ-universality rather than QQ-universality.

THEOREM 3.23 This is Observation 2 of [Nes+07]. Every CQ-universal resource can be made QQ-universal by allowing quantum inputs to be coupled into the resource state using the same coupling method, i.e. CZ gates.

We now introduce a new characterization of the universality of graph states.

DEFINITION 3.24 Let **LO** denote local operations, i.e. any single qubit unitary and measurement. Let **LC** denote local Clifford operations, i.e. any ZX diagram with spiders of arity at most 2 and phases multiples of $\frac{\pi}{2}$. Let **CC** denote **classical communication**, which represents the measurement correction and Pauli error tracking done on-the-fly in MBQC.

States $|\psi\rangle$ and $|\phi\rangle$ both on n qubits are called **LU-equivalent** if there is a unitary $U = \otimes_{i=1}^n U_i$, where every U_i is a local unitary, such that $|\psi\rangle = U|\phi\rangle$. In this case, write $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Now, if every U_i is made up of only local Clifford gates, say they are **LC-equivalent** and write $|\psi\rangle =_{\text{LC}} |\phi\rangle$.

THEOREM 3.25 This is the main result of [VDD04]. Graph states $|G_1\rangle$ and $|G_2\rangle$ are **LC-equivalent** iff there is a series of local complementations that transform one graph to another.

Proof sketch. The backwards direction follows from Lemma 2.29. The forward direction surmounts to showing that the successive application of local complementation generates the orbit of any graph state under local Clifford operations. Informally, the proof in [VDD04] does this by a clever characterization of local Clifford operations as graph operations, before showing that all these operations can be implemented using local complementation alone. $\square$

THEOREM 3.26 This is the main result of [Ji+08]. There are graphs that are **LU-equivalent** but not **LC-equivalent**. On the other hand, there are some types of graphs which are **LU-equivalent** iff they are **LC-equivalent** [GN07; VDD05].

Proof. In [Ji+08], they present 27-vertex graphs that are **LU-equivalent** but not **LC-equivalent**. It was recently shown in [Cla26] that this counterexample is minimal in the number of qubits. $\square$

THEOREM 3.27 A family of states Φ is a universal resource for MBQC if for all n -qubit states $|\psi\rangle$, $\exists |\phi\rangle \in \Phi$ with $m \geq n$ qubits, such that the transformation $|\phi\rangle \rightarrow |\psi\rangle$ is possible deterministically with **LO** and **CC**. We denote this by $|\phi\rangle \geq_{\text{LOCC}} |\psi\rangle$.

Proof. This follows almost immediately from the definition. $\square$

DEFINITION 3.28 Let Φ and Ψ be families of resource states. We say that Φ and Ψ are **LU-equivalent**, and write $\Phi =_{\text{LU}} \Psi$, if the following holds:

1. For every $|\phi\rangle \in \Phi$, there exists $|\psi\rangle \in \Psi$ such that $|\phi\rangle =_{\text{LU}} |\psi\rangle$, and
2. for every $|\psi\rangle \in \Psi$, there exists $|\phi\rangle \in \Phi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$.

Define **LC-equivalent** correspondingly.

LEMMA 3.29 Let Φ and Ψ be **LU-equivalent** families of resource states. Then Φ is a universal resource for MBQC if and only if Ψ is a universal resource for MBQC.

Proof. Suppose Φ is universal, and let $|\rho\rangle$ be any n -qubit target state. By universality, there exists $|\phi\rangle \in \Phi$ on $m \geq n$ qubits such that $|\phi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Since $\Phi =_{\text{LU}} \Psi$, there exists $|\psi\rangle \in \Psi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Local unitaries are reversible LOCC operations, so $|\psi\rangle \geq_{\text{LOCC}} |\phi\rangle$. Composing LOCC transformations gives $|\psi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Thus Ψ is universal. The reverse implication is identical. $\square$

Note that, if we have a family of resource states Φ that is not LU equivalent to some universal family of resource states Ψ , this does not necessarily imply Φ is not universal. This concludes our overview of the main universality results concerning LU-, LO, and LC equivalence. To finish off this section, we present one other universality criteria.

THEOREM 3.30 [From [Van+06]] Let Ψ be a family of resource states and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;
2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.

Then Ψ cannot be a universal resource.

Proof. The first condition states that LOCC cannot increase this measure, while the second says that the supremum value of E , which can be ∞ , is not reached among all $|\psi\rangle \in \Psi$. If both were true, then clearly universality is not possible. $\square$

The above theorem is very general and consequently very powerful, as we will soon see. However, note that satisfying the converse of the second condition for some measure E is not sufficient to guarantee universality. For example, the family of *toric code states* have unbounded entanglement width but are not universal [BR07].

THEOREM 3.31 Consider the family of all star graphs **Star**. A **star graph** on n vertices is a tree where one vertex has degree $n - 1$ and all other vertices have degree 1. Then **Star** is not a universal resource.

We choose to prove a more powerful result using rank-width as our measure E .

DEFINITION 3.32 (RANK WIDTH) Let $G = (V, E)$ be a graph. For any subset $A \subseteq V$, interpret it as a graph cut with A on one side and $V \setminus A$ on the other. Define the **cut-rank** of A by

$$\rho_G(A) := \text{rank}_{\mathbb{F}_2} M_A,$$

where M_A is the $|A| \times |V \setminus A|$ matrix whose entries are

$$(M_A)_{uv} = \begin{cases} 1 & \text{if } u \in A, v \in V \setminus A, \text{ and } uv \in E, \\ 0 & \text{otherwise.} \end{cases}$$

In words, $\rho_G(A)$ is the rank, over $\mathbb{F}_2$, of the adjacency matrix recording which edges cross the cut

$$A \mid V \setminus A.$$

A **rank decomposition** of G is a binary tree T whose leaves are the vertices of G . Every edge e of T splits the leaves into two parts, and therefore determines a cut

$$A_e \mid V \setminus A_e$$

of the vertices of G . The **width** of the decomposition T is

$$\max_{e \in E(T)} \rho_G(A_e).$$

Finally, the **rank width** of G is the smallest possible width over all such decompositions:

$$\text{rwd}(G) := \min_T \max_{e \in E(T)} \rho_G(A_e).$$

Thus, rank width measures how well the vertices of G can be arranged in a tree so that every cut appearing in the tree has low binary adjacency rank.

LEMMA 3.33 Every tree has rank width at most 1.

Proof. Let $G = (V, E)$ be a tree. We will construct a rank decomposition where every cut has cut-rank at most 1. Root the tree G at some vertex r . For each vertex $v \neq r$, let G_v denote the subtree consisting of v and all of its descendants.

Let $A_v := V(G_v)$. Since G is a tree, there is exactly one edge between A_v and $V \setminus A_v$, namely the edge from v to its parent. Therefore, the cut matrix M_{A_v} has exactly one nonzero entry. Hence

$$\rho_G(A_v) = \text{rank}_{\mathbb{F}_2} M_{A_v} \leq 1.$$

Now choose a binary decomposition tree T that recursively separates rooted subtrees of G . Every cut appearing in this decomposition is of the form $A_v \mid V \setminus A_v$ or a union of such subtrees attached to a common vertex. In either case, because G is a tree, the edges crossing the cut are incident to a single vertex on one side of the cut. Thus the corresponding bipartite adjacency matrix has all nonzero entries contained in a single row or a single column, so its rank over $\mathbb{F}_2$ is at most 1. Hence this rank decomposition has width at most 1. Therefore $\text{rwd}(G) \leq 1$. $\square$

Proof of Theorem 3.31. Every graph in **Star** is a tree, so by Lemma 3.33 any Star graph has at most rank width 1. On the other hand, C_5 , the cycle on 5 vertices, has rank-width at least 2. To see this, take any rank decomposition of C_5 . This tree has 5 leaves, so there must be some edge whose removal separates the leaves into a $2 \mid 3$ split. It suffices to show that every $2 \mid 3$ cut of C_5 has cut-rank 2.

Let C_5 have vertices labelled by $[5]$. Let $A = \{u, v\}$ for some distinct $u, v \in V(C_5)$ and $B = V(C_5) \setminus A$. If u and v are adjacent, by symmetry it suffices to consider $A = \{1, 2\}$. Then

$$M_A = \begin{array}{c|ccc} & 3 & 4 & 5 \\ \hline 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{array}. \quad (3.34)$$

Hence the cut-rank of A is 2. Now if u and v are not adjacent, by symmetry it suffices to consider $A = \{1, 3\}$. Then

$$M_A = \begin{array}{c|ccc} & 2 & 4 & 5 \\ \hline 1 & 1 & 0 & 1 \\ 3 & 1 & 1 & 0 \end{array}. \quad (3.35)$$

Thus the cut-rank of A is 2. Hence $\text{rwd}(C_5) \geq 2$. Theorem 3.30 completes the proof. $\square$

THEOREM 3.34 Local complementing a graph G at any vertex does not change its rank-width.

Proof. The overall goal is to show that local complementation only performs elementary row and column operations on every cut matrix, so it preserves the cut-rank. For a cut $A \mid B$ of a graph G , let $M_A(G) = M_A$ from Definition 3.32. Then we wish to show

$$\text{rank}_{\mathbb{F}_2} M_A(G) = \text{rank}_{\mathbb{F}_2} M_A(G \star v) \quad (3.36)$$

for all cuts $A \mid B$ and $v \in V(G)$. Suppose first that $v \in A$. Let r_v be the row of $M_A(G)$ indexed by v . Let $u \in A \setminus \{v\}$. Local complementation at v toggles the edge uw , with $w \in B$, exactly when both u and w are adjacent to v . Then the row of $M_A(G \star v)$ indexed by u is

$$r'_u = \begin{cases} r_u + r_v & \text{if } uv \in E(G), \\ r_u & \text{if } uv \notin E(G), \end{cases}$$

where the addition is over $\mathbb{F}_2$. The row indexed by v is unchanged, since local complementation does not change edges incident to v . This is an elementary row operation.

Now suppose $v \in B$. The same argument applies with columns instead of rows. Let c_v be the column of $M_A(G)$ indexed by v . For each $w \in B \setminus \{v\}$, local complementation at v toggles the edge uw , with $u \in A$, exactly when both u and w are adjacent to v . Thus the column indexed by w becomes

$$c'_w = \begin{cases} c_w + c_v & \text{if } vw \in E(G), \\ c_w & \text{if } vw \notin E(G). \end{cases}$$

The column indexed by v is unchanged. Once again, this is an elementary column operation. Therefore every cut has the same cut-rank in G and in $G \star v$. Thus $\text{rwd}(G) = \text{rwd}(G \star v)$. $\square$

THEOREM 3.35 The family **KGraph** of complete graph states is not universal.

Proof. We first show that every complete graph is local-complementable to a star. Let n be the number of vertices. $n = 1$ is trivial. For $n \geq 2$, local complementation at any $v \in V(K_n)$ toggles all edges not incident to v . The result is a star rooted at v . By Theorem 3.34 and Theorem 3.31, **KGraph** is not universal. $\square$

4 The Comparability Grid

4.1 Circle Graphs

DEFINITION 4.1 The $m \times n$ **comparability grid** is a graph $G = (V, E)$ with

$$V = \{(i, j) \mid i \in [m], j \in [n]\} \quad (4.1)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, i_2 \geq i_1, j_2 \geq j_1\} \} \quad (4.2)$$

In other words, m rows and n columns with edges defined by the partial order $\leq$. Denote this graph as $\text{CGrid}_{m \times n}$. Let CGrid denote the family of all such graphs.

EXAMPLE 4.2 The figure below shows the 3×3 comparability grid:

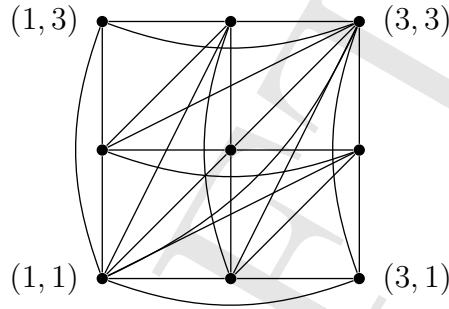


Figure 4: $\text{CGrid}_{3 \times 3}$.

The comparability grid is the “maximal” object of what are called circle graphs.

DEFINITION 4.3 A **circle graph** is an intersection graph of chords of a circle. Equivalently, a graph $G = (V, E)$ is a circle graph if we can assign each $v \in V$ a chord c_v in a circle such that

$$uv \in E \iff c_u \text{ and } c_v \text{ intersect.} \quad (4.3)$$

Denote the family of all circle graphs as CGraph .

To see that comparability graphs are circle graphs, we introduce another kind of circle graph: permutation graphs.

DEFINITION 4.4 A **permutation graph** is a graph that records which pairs of objects are in opposite order in two different lists. More precisely, let V be a finite set of labels, and choose two total orders $<_1$ and $<_2$ on V . The permutation graph $(G = V, E)$ associated with these two orders has V as its vertices, and edges

$$uv \in E \iff (u <_1 v \wedge v <_2 u) \vee (v <_1 u \wedge u <_2 v).$$

To draw a permutation graph, place the vertices in $<_1$ order on one horizontal line, and place the vertices in $<_2$ order on a second horizontal line below it. Connect matching labels with a line segment. Then two vertices are adjacent exactly when their line segments cross. Denote the family of all permutation graphs as PGraph .

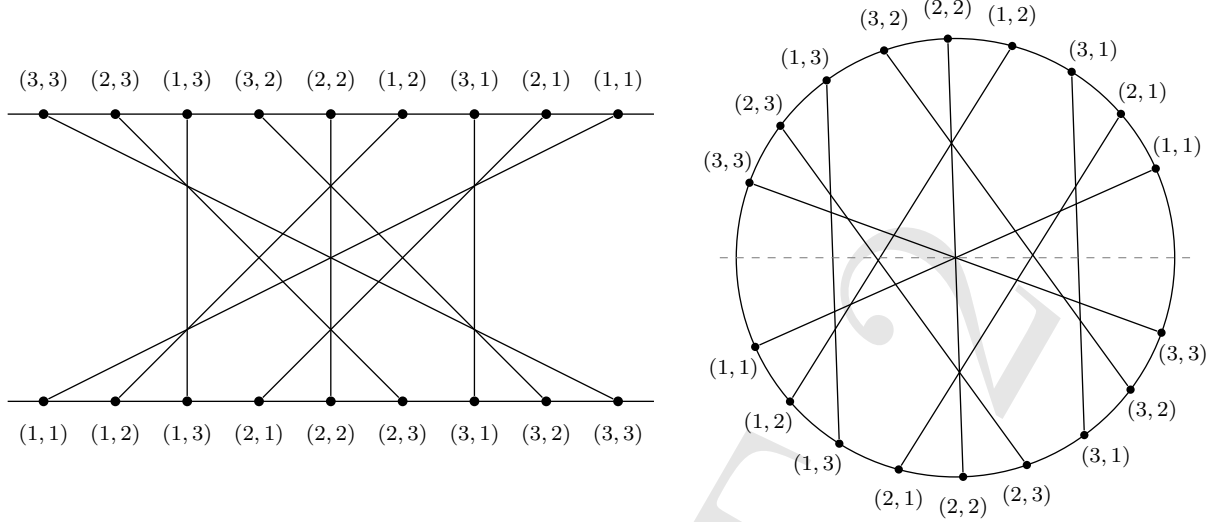


Figure 5: A permutation graph representation of $\text{CGrid}_{3 \times 3}$ and the equivalent circle graph representation.

THEOREM 4.5 A comparability grid is a permutation graph, and hence a circle graph.

Proof. Fix a $G = \text{CGrid}_{m \times n}$. We start by recasting the edge condition to a more tractable form. For vertices $(i, j), (i', j') \in V(G)$, write $(i, j) \sim_G (i', j')$ if there is an edge between them in G . Then

$$(i, j) \sim_G (i', j') \iff (i - i')(j - j') > 0.$$

Define two orders on $V(G)$:

$$(i, j) <_1 (i', j') \iff i < i' \vee (i = i' \wedge j < j') \quad (4.4)$$

$$(i, j) <_2 (i', j') \iff j > j' \vee (j = j' \wedge i > i') \quad (4.5)$$

These are lexicographic orders and thus total orders. Now, draw a permutation diagram with vertices on the top line in order $<_2$ and $<_1$ on the bottom line. Connect equal labels with a line segment. Two segments cross exactly when their two labels appear in opposite orders on the two lines. This happens exactly when $(i - i')(j - j') \geq 0$. Hence G is a permutation graph.

Finally, we show that any permutation graph is a circle graph. Place the two parallel lines of a permutation graph inside a circle and identify their endpoints along the boundary, such that each line segment becomes a chord of a circle. This preserves crossings. Therefore every permutation graph is a circle graph. $\square$

THEOREM 4.6 This is Theorem 1 of [Hah+26]. For circle graph states, $\text{LU} = \text{LC}$. Furthermore, the only graphs LU -equivalent to circle graph states are circle graphs themselves. In other words, CGraph is closed under local complementation.

DEFINITION 4.7 A graph H is a **vertex minor** of a graph G if there a sequence of vertex deletions and local complementations from G to H .

THEOREM 4.8 Circle graphs are closed under vertex minors.

Proof. This follows from Theorem 4.6. □

THEOREM 4.9 This is Corollary 53 of [BT16], strengthened by [Hah+26]. Every circle graph is a vertex minor of some bipartite circle graph that is at most quadratically larger in the number of vertices.

THEOREM 4.10 This is Lemma 3.1, 3.2, and 3.3 of [Gee+23a]. Every circle graph on n vertices is isomorphic to a vertex minor of the $3n \times 3n$ comparability grid. More specifically, every n -vertex permutation graph is isomorphic to an induced subgraph of the $n \times n$ comparability grid, and every circle graph on n vertices is a vertex minor of a permutation graph on $3n$ vertices.¹⁰

THEOREM 4.11 This is Corollary 3 and 4 of [Hah+26]. MBQC on circle graph states is efficiently classically simulable. Furthermore, assuming $\text{BPP} \neq \text{BQP}$ ¹¹, polynomial rank width is not a sufficient condition for efficient universality of a graph state resource MBQC.

The previous theorem begins to highlight connections between rank-width and the circle graphs. Restricting our attention to comparability grids, we observe even more connections.

THEOREM 4.12 The rank-width of the $n \times n$ comparability grid is at least $n/4$. Thus, CGrid has unbounded rank-width, and so does PGraph and CGraph.

Proof. The first claim is Lemma 4 of [Hah+26]. The second follows by Theorem 4.5. □

THEOREM 4.13 This is Theorem 1.1 and 1.3 of [Gee+23b]. There is a function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ so that for every positive integer n , every graph of rank-width at least $f(n)$ has a vertex minor isomorphic to the $n \times n$ comparability grid. On the other hand, there is a function $g : \mathbb{Z} \rightarrow \mathbb{Z}$ so that for every circle graph H , every graph with no vertex-minor isomorphic to H has rank-width at most $g(|V(H)|)$.

We conclude this section by contrasting the above results for circle graphs and comparability grids with a result for grid graphs.

THEOREM 4.14 This is Theorem 12 and 19 of [Cam+24]. Every planar graph is a vertex minor of some grid. Furthermore, every graph is a vertex minor of some planar graph. Thus every graph is a vertex minor of some grid.

¹⁰This is what we meant when we introduced comparability grids as the “maximal” object of circle graphs in the remark before Definition 4.3.

¹¹BPP is the classical complexity class for bounded polynomial-time, defined in [Gil77]. BQP is the quantum complexity class for bounded-error quantum polynomial-time, defined in [BV93]

This is in sharp contrast with the following result.

THEOREM 4.15 This is the main result of [Bou94]. A graph is a circle graph if and only if no graph obtained by a sequence of local complementations to G has an induced subgraph isomorphic to one of the graphs below. These are known as the **forbidden minors** of circle graphs (and consequently, by Theorem 4.10, of comparability grids).

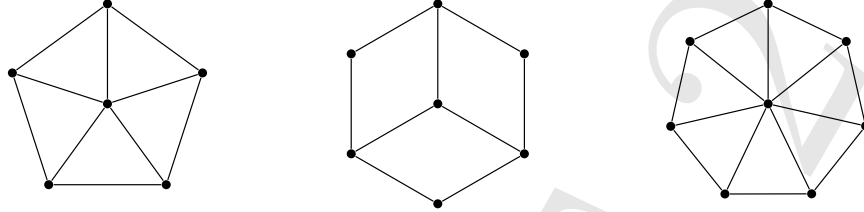


Figure 6: Forbidden vertex minors of circle graphs.

4.2 Subgraphs of the Comparability Grid

Recall that, by Fact 3.15, vertex deletion is an operation allowable in the Measurement Calculus model of MBQC. Thus, if we can show that a family of induced subgraphs of comparability grids is universal, we obtain that comparability grids are universal. We begin by characterizing the data of a comparability grid.

FACT 4.16 Let H be an induced subgraph of a comparability grid CGrid , with $|V(H)| = n$. Then H admits a canonical encoding by a permutation $\pi(H) \in S_n$. Conversely, every permutation $\pi \in S_n$ determines an induced subgraph $H(\pi)$ of CGrid .

Proof. Write the vertices of H as points (x, y) in the comparability grid, where two vertices are adjacent precisely when they are comparable:

$$(x_1, y_1)(x_2, y_2) \in E(H) \iff (x_1 - x_2)(y_1 - y_2) > 0. \quad (4.6)$$

Define a perturbed point set

$$V' := \left\{ \left(x + \frac{y}{n+1}, y + \frac{x}{n+1} \right) : (x, y) \in V(H) \right\}. \quad (4.7)$$

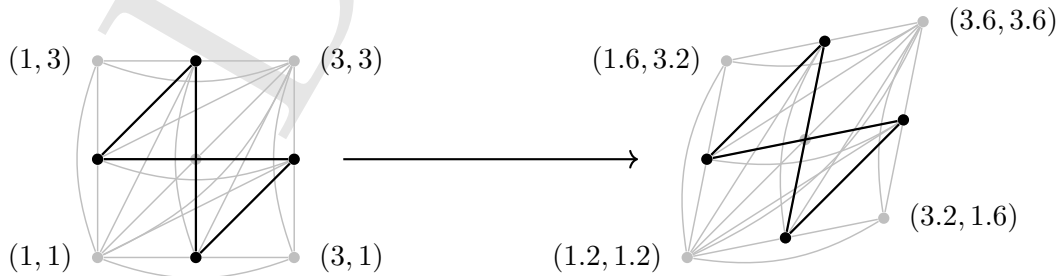


Figure 7: The perturbation of $H(2143)$ in a $\text{CGrid}_{3 \times 3}$.

This perturbation preserves the relative comparability of every pair of vertices, but ensures that no two vertices have the same first coordinate and no two vertices have the same second coordinate. Indeed, if

$$x_1 + \frac{y_1}{n+1} = x_2 + \frac{y_2}{n+1} \Rightarrow x_1 - x_2 = \frac{y_2 - y_1}{n+1}. \quad (4.8)$$

The left-hand side is an integer, while the right-hand side has absolute value strictly less than 1. Hence both sides are 0, so $x_1 = x_2$ and $y_1 = y_2$. Thus the vertices are identical. The same argument applies to the second coordinates.

Now sort the perturbed vertices by increasing first coordinate. Since the second coordinates are all distinct, their relative order is a permutation of $[n]$. Denote this permutation by $\pi(H)$. Thus, after relabeling the vertices by their x -order, we may write the vertices as

$$v_i = (i, \pi(H)(i)), \quad i \in [n]. \quad (4.9)$$

The adjacency rule becomes

$$v_i v_j \in E(H) \iff (i - j)(\pi(H)(i) - \pi(H)(j)) > 0. \quad (4.10)$$

Hence H is exactly the permutation graph encoded by $\pi(H)$.

Conversely, given $\pi \in S_n$, define

$$V(H(\pi)) := \{(i, \pi(i)) : i \in [n]\}, \quad (4.11)$$

with the induced comparability grid edges. This gives an induced subgraph $H(\pi) \subseteq \mathbf{CGrid}$. $\square$

FACT 4.17 Two induced subgraphs H, H' of a comparability grid may satisfy $\pi(H) \neq \pi(H')$ while still being isomorphic as abstract graphs.

Proof. Consider the permutations

$$\pi = 132, \quad \sigma = 213. \quad (4.12)$$

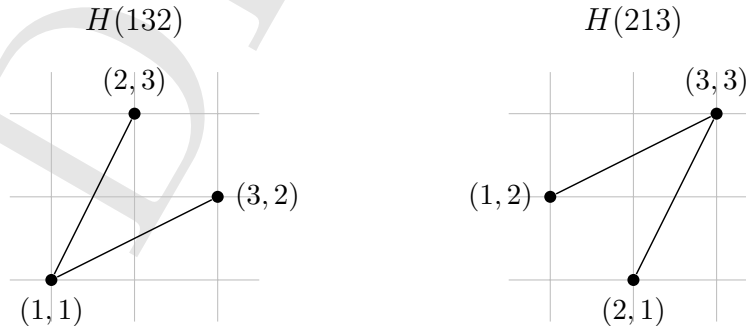


Figure 8: The induced comparability-grid subgraphs $H(132)$ and $H(213)$.

Clearly $H(\pi)$ is isomorphic to $H(\sigma)$ though $\pi \neq \sigma$. $\square$

FACT 4.18 Let $\pi \in S_n$. Define its **reverse-complement** by

$$\pi^{\text{rc}}(i) := n + 1 - \pi(n + 1 - i). \quad (4.13)$$

Then $H(\pi) \cong H(\pi^{\text{rc}})$.

Proof. Write the vertices of $H(\pi)$ as $v_i = (i, \pi(i))$ for $i \in [n]$. Define

$$f(v_i) := (n + 1 - i, n + 1 - \pi(i)). \quad (4.14)$$

As i ranges over $[n]$, the image is exactly

$$\{(k, \pi^{\text{rc}}(k)) : k \in [n]\}, \quad (4.15)$$

so f is a bijection from $V(H(\pi))$ to $V(H(\pi^{\text{rc}}))$.

For $i \neq j$, adjacency in a comparability grid is given by

$$v_i v_j \in E(H(\pi)) \iff (i - j)(\pi(i) - \pi(j)) > 0. \quad (4.16)$$

Under f , both differences change sign:

$$(n + 1 - i) - (n + 1 - j) = -(i - j), \quad (4.17)$$

$$(n + 1 - \pi(i)) - (n + 1 - \pi(j)) = -(\pi(i) - \pi(j)). \quad (4.18)$$

Therefore the product

$$(i - j)(\pi(i) - \pi(j)) \quad (4.19)$$

is preserved. Hence $v_i v_j$ is an edge if and only if $f(v_i) f(v_j)$ is an edge. Thus f is a graph isomorphism, and $H(\pi) \cong H(\pi^{\text{rc}})$. $\square$

This representation of a subgraph will come very handy in Section 5. We now conclude this section with a few observations regarding whether grids are induced subgraphs of some comparability grid.

FACT 4.19 For all $n \in \mathbb{N}$, $\text{Grid}_{2 \times n}$ is a subgraph of a $\text{CGrid}_{n+1 \times n+1}$.

Proof. Repeat and “untwist” the following construction: $\square$

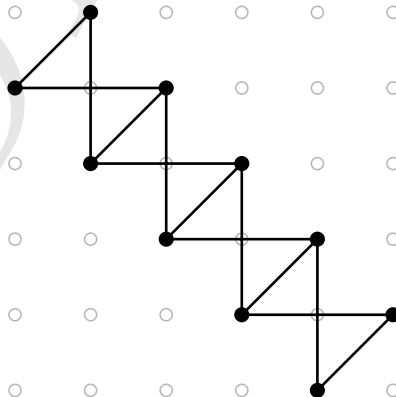


Figure 9: A $\text{Grid}_{2 \times 5}$ in a $\text{CGrid}_{6 \times 6}$.

This result is quite exciting, as it immediately implies that **CGrid** is 2-qubit universal, as the family of all $\text{Grid}_{2 \times n}$ is 2-qubit universal. Unfortunately, it turns out that we cannot do much better.

FACT 4.20 $\text{Grid}_{3 \times 4}$ is not a vertex minor of any circle graph. Since vertex minors is a strictly more powerful operation than vertex deletion, $\text{Grid}_{3 \times 4}$ is not an induced subgraph of any **CGrid**.

Proof. Look at the following figure:

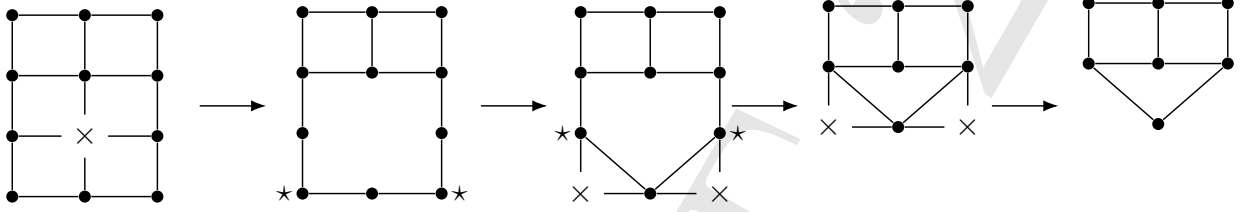


Figure 10: Vertices marked by $\star$ are locally complemented, and vertices marked by $\times$ are deleted. This shows that $\text{Grid}_{3 \times 4}$ has a forbidden minor of circle graphs as a vertex minor.

The rightmost graph is a forbidden vertex minor of circle graphs. If the $\text{Grid}_{3 \times 4}$ was the vertex minor of some circle graph, then so would be this forbidden minor. Thus, no circle graph, and consequently no comparability grid, has any $\text{Grid}_{m \times n}$ with $m \geq 3$ and $n \geq 4$ as a vertex minor. $\square$

An exhaustive search (using the algorithms described in the next section) reveals that we may strengthen this result a little bit.

FACT 4.21 $\text{Grid}_{3 \times 3}$ is not a induced subgraph of any circle graph. Consequently, $\text{Grid}_{m \times n}$ is not an induced subgraph of any circle graph, for $m, n \geq 3$.

Proof. The proof surmounts to enumerating every possible 9-vertex subgraph of a **CGrid**, and concluding that none are isomorphic to $\text{Grid}_{3 \times 3}$. $\square$

Let us summarize what we have seen this section. Through vertex minors, we see that any potential proof of the universality of the comparability grid cannot involve reducing, via local complementation and vertex deletion, a comparability grid into some desired grid graph. This is the implication of Fact 4.20.

Furthermore, if the comparability grid is universal, its MBQC protocol must asymptotically use a strictly super-polynomial or faster growing number of qubits with respect to the number of qubits in the desired output quantum state. This follows from 4.11.

Suppose comparability grids are universal. If we are to present a constructive proof of this, i.e. given a target state $|\phi\rangle$ on n qubits, an explicit vertex minor of some comparability grid and a measurement pattern that generates $|\phi\rangle$ using it, then this vertex minor must have asymptotically super-polynomial or larger number of vertices with respect to n . In the next section, we outline the key methodology of this thesis: a series of algorithms for enumerating and characterizing the vertex minors of comparability graphs. We run these algorithms to create constructions which we conjecture may show the universality of the comparability grid. These are presented in Section 6.

5 Algorithmic Approaches

5.1 Overview

This section is unfinished, but it wishes to introduce the following items: In Section 5.2, we will show that using the permutation representation, we may enumerate induced subgraphs of the comparability grid. This allows us to answer questions like “Is the 3 by 3 Grid every an induced subgraph of some comp grid?” Then, we will show how to, given a graph, determine if it is an induced subgraph of the comparability grid. The way we do this is by encoding the graph’s data as a linear program, and checking if the LP is satisfiable. We conclude the section with a discussion to determine if known subgraph of the comp grid is tileable. This can be done by tiling the subgraph, and then using the LP code to check if it is still a subgraph. The LP code is constructive, so it will also output the construction if this is true. We remark that determining if a given graph is a vertex minor of some cgrid is an extremely expensive operation - it is NP-complete to check if its the vertex minor of a specific cgrid.

In section 5.3, we will introduce the idea of gadget extraction. Namely, give a subgraph that we know is tileable, how can we extract the circuit gadget we implement. We start with the simple example of a 3 wide grid with one additional edge every column, using purely circuit representation. We then show its ZX derivation, and say that because of the theorems in Section 2, we can find the gadget of any tileable subgraph efficiently. Now, given a Clifford circuit, can we determine if there is a graph state that implements it? I need a bit more time, but this may be possible using ZX.

In section 5.4, we will show the Lie algebra formalism for determining if a k-qubit gadget is k-qubit universal. There are some new developments in this area that I am excited to share, but this is mostly a copy-paste of the Lie Algebra notes Professor Grier made.

We then transition to the next section, where we “present the fruits of our labor,” or really in this case, the “fruits of the labor of the programs.”

5.2 Subgraph Algorithms

5.3 Gadget Extraction

In this section, we will focus on how we may extract a circuit gadget that captures the computational power of a tileable subgraph of a CGrid.

FACT 5.1 We start with a useful identity:

$$\begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \bullet \\ | \\ \oplus \end{array} \begin{array}{c} \oplus \\ | \\ \bullet \end{array} = \begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \oplus \\ | \\ \oplus \end{array} \begin{array}{c} \oplus \\ | \\ \oplus \end{array} \quad (5.1)$$

Proof. Note that

$$(5.2)$$

This completes the proof. $\square$

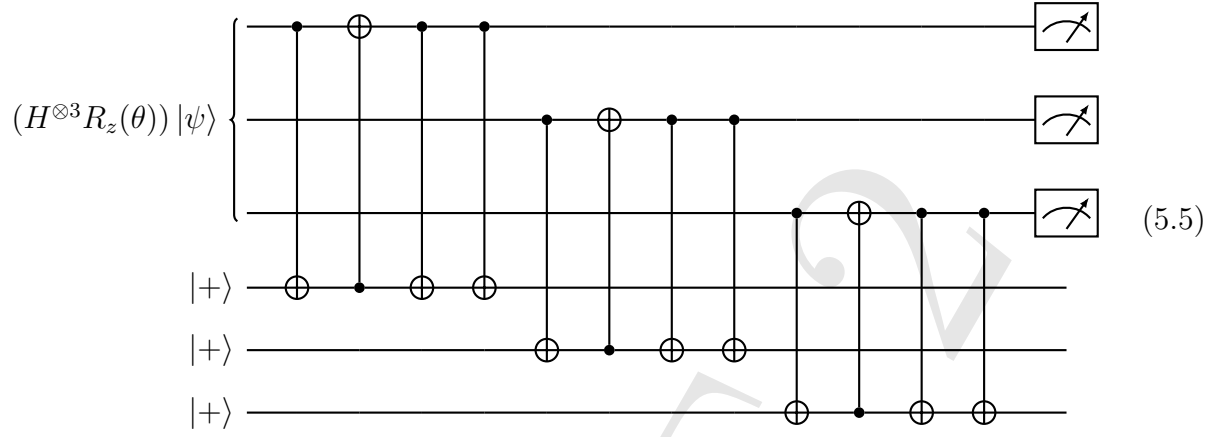
DEFINITION 5.2 We define a gadget that captures the computational power of the construction in Definition ?? . Start with the standard MBQC protocol:

$$(5.3)$$

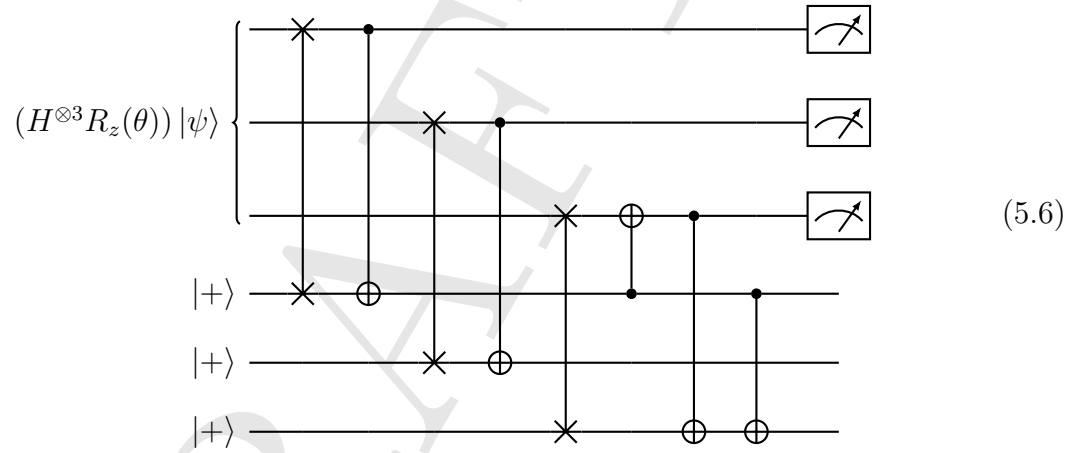
Pushing the gates through like in Equation 3.2:

$$(5.4)$$

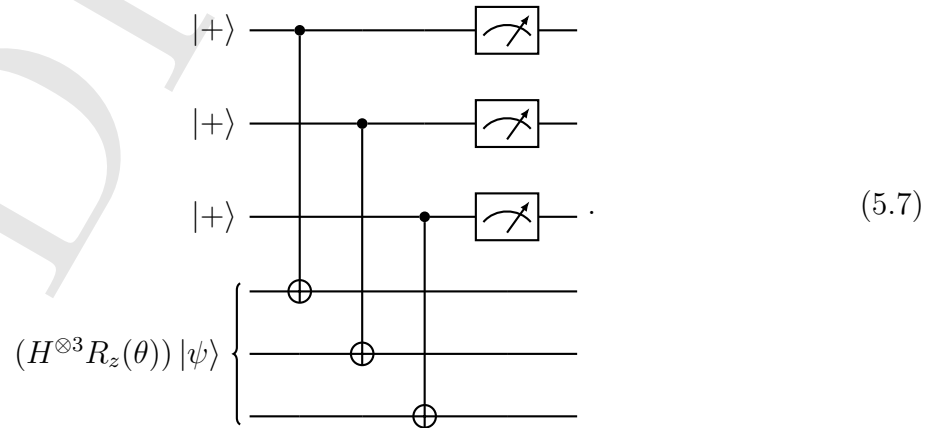
We can add the following CNOTs with affecting the circuit, like in Equation 3.3:



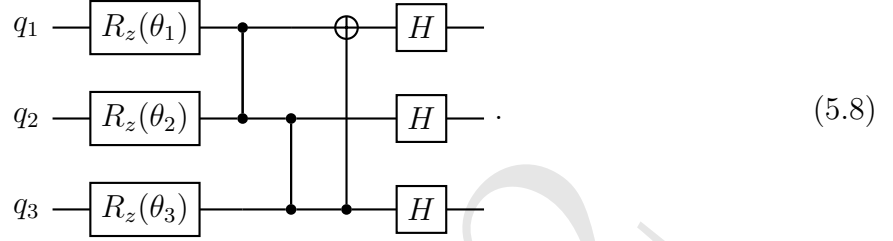
Applying the SWAP gate identity from Equation 3.4:



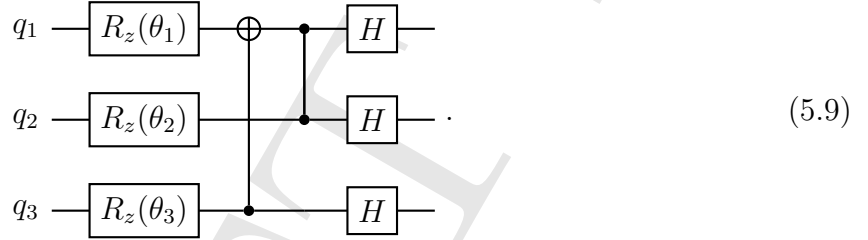
Finally, commuting the swap gates to the beginning gives



Dropping the Pauli gates, we get that we can apply blocks of



Call the above the **width-3 gadget**. Using Fact 5.1, the gadget simplifies to



THEOREM 5.3 [SK17] provides an algorithm for determining the universality of a gate set, in terms of a linear algebra problem. Though it is not polynomial in terms of the number of qubits and gates, it is massively parallelizable. Ideally, I would like a GPU kernel that does much of this work.

5.4 Lie Algebra Computations

6 Conclusions

6.1 Conjectures

This is unfinished, but here is the main content: there are two iterative constructions which pass the Lie Algebra test up to 8 qubits. The first is the middle-all-connected construction: let k be the number of qubits we desire for our computation. $k = 1$ is trivial. $k = 2$ is the $\text{Grid}_{2 \times n}$.

To construct the graph state for $k+1$, take two copies of k and tile them together. Assume the direction of MBQC is rightwards. Add three new vertices: One on the anti-diagonal of the first column, one on the middle column that joins the two copies of the k construction, and one on the rightmost column. Then these three new vertices, v_1 , v_2 , and v_3 respectively, form a complete graph with all other vertices in the same column / anti-diagonal.

Now, to make this construction an induced subgraph of some CGrid , it turns out it suffices to make v_2 connected to all other vertices, including v_1 and v_3 . This construction, when converted to its corresponding circuit and is tested with the Lie algebra criterion, outputs universal up to $k = 8$. This takes about 15 minutes to run on a standard laptop.

The other construction is called all-connected. This is similar, but now have v_1 also connect to every vertex on the left copy of the k construction. Then have v_3 connect to every vertex on the right copy of the k construction. This turns out to be universal up to 8 qubits too.

Remark. 8 is such a large number that we may conjecture that this holds, for both constructions, for all k . Note that the problem becomes exponentially more difficult for every additional k we wish to verify computationally. This is because the size of the gadget increases exponentially, and so does the size of the Pauli basis we wish to completely generate.

Furthermore, a simple inductive proof is almost certainly impossible. For one, both $k+1$ constructions have many edges into all the vertices of the two k gadgets within it. This makes it hard to use any facts about the k gadget. Furthermore, it is not necessarily true that the k gadget is one-step universal; we may need to tile the k gadget many, many times to make it universal for k qubits.

Now suppose we have a construction where, by some miracle, in the $k+1$ construction we may choose to apply the k gadget as many times as we want. An easy proof shows that if we do this, there *must* be additional edges from the $k+1$ th qubit into each of these k gadgets. In other words, a simple inductive proof where we say "because the k th gadget is universal for k qubits, it can handle any after effects of the $k+1$ th qubit" seem intractable.

Given the theory we have developed, there are five ways we may move forward.

First, we need to incorporate local complementation into every part of our construction. It may be possible that there is an iterative construction like the one above that, due to local complementation, become significantly easier to construct and prove theoretically.

Second, we should understand more precisely what made the all-connected and middle-all-connected constructions work.

Third, we may attempt to more coarsely characterize the problem. Consider the set of all grid graphs with k additional edges. For small graphs, we may compute the number of Pauli bases this gadget generates, taking for granted that the construction is tileable. Now create a plot, whose x axis is the number of additional edges k , and whose y axis is the

average number of Pauli bases generated by this family with k additional edges. What does this distribution look like?

Fourth, consider instead a random variable that outputs a random vertex minor of the comparability grid that is guaranteed to have some grid as a subgraph. Can we use probabilistic methods to force a universality result out of this?

Fifth, we have seen how vertex minors and rank-width are very powerful concepts for characterizing and studying the universality of comparability grids. What other measures are there?

6.2 Future Directions

Random measure idea.

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DRAFT 2

A Appendix: Background on Quantum Computing

A.1 Qubits

DEFINITION A.1 (QUBIT) A single **qubit** $|\phi\rangle$ is a state

$$|\phi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{C}^2, \quad (\text{A.1})$$

where $|\alpha|^2 + |\beta|^2 = 1$ and $\alpha, \beta \in \mathbb{C}$. We say $|\phi\rangle$ is in a **superposition** between $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Thus a single qubit is a length 2 vector in the Hilbert space $\mathbb{C}^2$ with a ℓ_2 norm of 1.

DEFINITION A.2 (BRAKET NOTATION) Define “kets” $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, both in $\mathbb{C}^2$. For any $|\phi\rangle$, let “bra” $\langle\phi| := |\phi\rangle^\dagger = (|\phi\rangle^*)^T$ be its conjugate transpose. Thus the braket denotes the inner product:

$$\langle\phi|\phi\rangle := \langle\phi| \cdot |\phi\rangle. \quad (\text{A.2})$$

Now the ℓ_2 norm of $|\phi\rangle$ can now be written as $\| |\phi\rangle \|_2 = \sqrt{\langle\phi|\phi\rangle}$.

DEFINITION A.3 (TENSOR PRODUCT) Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{o \times p}$. For our purposes, it suffices to define the **tensor product**

$$A \otimes B := \begin{bmatrix} a_{1,1}B & \cdots & a_{1,n}B \\ \vdots & \ddots & \vdots \\ a_{n,1}B & \cdots & a_{n,m}B \end{bmatrix} \quad (\text{A.3})$$

where $a_{i,j}$ denotes the i th row and j th column of A .

DEFINITION A.4 (MULTIPLE QUBITS) Let $x = \overline{x_1 x_2 \cdots x_d} \in \{0, 1\}^d$, i.e. a binary bit-string. Define

$$|x\rangle := \bigotimes_{i=1}^d |x_i\rangle. \quad (\text{A.4})$$

For example, $|00\rangle = |0\rangle \otimes |0\rangle = (1 \ 0 \ 0 \ 0)^T$. We may also replace x by its integer evaluation as a binary number, by abuse of notation. For example,

$$|3\rangle = |11\rangle = (0 \ 0 \ 0 \ 1)^T \quad (\text{A.5})$$

but we may also have

$$|3\rangle = |011\rangle = (0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0)^T \quad (\text{A.6})$$

Which binary representation we choose should be clear from context. For example, this gives us two equivalent definitions of a q -qubit state

$$|\varphi\rangle = \sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = \sum_{x \in [2^d] \cup \{0\}} \alpha_x |x\rangle \in \mathbb{C}^{2^d} \quad (\text{A.7})$$

where $\alpha_x \in \mathbb{C}$ and $\sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = 1$. As the number of dimensions increases exponentially with regards to the number of qubits, bracket notation will become increasingly concise compared to traditional vector notation.

DEFINITION A.5 (TENSOR PRODUCT NOTATION) For brevity, we will write $|\phi\rangle |\psi\rangle = |\phi\rangle \otimes |\psi\rangle$, so adjacent kets imply $\otimes$. However, note that $|\phi\psi\rangle \neq |\phi\rangle |\psi\rangle$ in general. Also, let

$$A^{\otimes n} := \bigotimes_{i=1}^n A. \quad (\text{A.8})$$

DEFINITION A.6 Vectors $|v_1\rangle, \dots, |v_n\rangle \in \mathbb{C}^n$ are an orthonormal basis if

1. $\|v_i\| = 1$ for every $i \in [n]$ and
2. $|v_i\rangle \perp |v_j\rangle$ (equivalently, $\langle v_i | v_j \rangle = 0$) for every $i, j \in [n]$ with $i \neq j$.

Henceforth, all bases are orthonormal unless explicitly said otherwise.

DEFINITION A.7 (MEASUREMENT) The **Born Rule** defines how quantum measurement occurs: An n -qubit state $|\phi\rangle$ can be measured under any orthonormal basis $\{|v_i\rangle\}_{i \in [2^n]}$ of $\mathbb{C}^{2^n}$. When measured, the output i occurs with probability $|\langle v_i | \phi \rangle|^2$. After seeing outcome i , the state collapses to $|v_i\rangle$. This process is irreversible.

DEFINITION A.8 (PARTIAL MEASUREMENT) One can also view measurement as projection. This is useful for **partial measurement**, where only some qubits in a state are measured. Let the n -qubit space $\mathbb{C}^{2^n} = S_1 \otimes S_2$, where $S_1 \cong (\mathbb{C}^2)^{\otimes k}$. Let $\{|s_i\rangle\}_{i \in [2^k]}$ be an orthonormal basis for S_1 . On outcome $i \in [2^k]$, define the **projection matrix** on S_1 as

$$P_i := |s_i\rangle \langle s_i|, \quad \text{such that} \quad P_i = P_i^\dagger, \quad P_i^2 = P_i. \quad (\text{A.9})$$

Define the **partial measurement** on S_1 by $M_i = P_i \otimes I \in \mathbb{C}^{2^n \times 2^n}$. For a state $|\phi\rangle \in \mathbb{C}^{2^n}$,

$$\Pr(\text{measuring } i \text{ from } |\phi\rangle) = \langle \phi | M_i | \phi \rangle = \|M_i |\phi\rangle\|^2. \quad (\text{A.10})$$

On measuring outcome i , the post-measurement state becomes

$$|\phi_i\rangle = \frac{M_i |\phi\rangle}{\sqrt{\langle \phi | M_i | \phi \rangle}}. \quad (\text{A.11})$$

EXAMPLE A.9 Consider the states

$$|+\rangle := \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \quad |-\rangle := \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle. \quad (\text{A.12})$$

Measuring either in the **standard basis** $\{|0\rangle, |1\rangle\}$ gives $|0\rangle$ with $\frac{1}{2}$ probability since

$$|\langle 0|+\rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 \right|^2 = \frac{1}{2} \quad (\text{A.13})$$

and similarly for $|1\rangle$. The squaring makes $|-\rangle$ proceed similarly as well.

DEFINITION A.10 A two qubit state $|\phi\rangle$ is **entangled** if it cannot be written as the tensor product of single qubits. That is, $\nexists |\psi\rangle, |\varphi\rangle \in \mathbb{C}^2$ such that $|\phi\rangle = |\psi\rangle \otimes |\varphi\rangle$. If a state is not entangled, it is called **separable**. An n qubit state $|\psi\rangle$ is entangled (or *contains* entanglement) if it is not fully separable.

A.2 Quantum Gates

DEFINITION A.11 A 1-qubit quantum gate U is a matrix $U \in \mathbb{C}^{2 \times 2}$ with:

1. $U^\dagger U = U U^\dagger = I$;
2. equivalently, U is a unitary matrix;
3. equivalently, $U = |\phi_0\rangle\langle 0| + |\phi_1\rangle\langle 1|$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ form an orthonormal basis.

Common single qubit gates are given in Table 2.

THEOREM A.12 (EULER DECOMPOSITION) Every one-qubit unitary U can be written as

$$U = e^{i\alpha} R_z(\beta) R_x(\gamma) R_z(\delta), \quad (\text{A.14})$$

where $\alpha, \beta, \gamma, \delta \in [0, 2\pi)$.

Proof. This is a standard proof, seen for example in Section 4.2 of [MikeIke]. Since U is unitary, $|\det U| = 1$. Choose $\alpha \in [0, 2\pi)$ such that $\det U = e^{2i\alpha}$. Let $V := e^{-i\alpha} U$ satisfy $\det V = 1$ while also being unitary. Now let

$$V = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad (\text{A.15})$$

Since the columns of V are orthonormal, the second column must be a unit vector orthogonal to the first. Thus

$$\begin{pmatrix} b \\ d \end{pmatrix} = \lambda \begin{pmatrix} -\bar{c} \\ \bar{a} \end{pmatrix} \quad (\text{A.16})$$

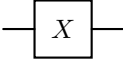
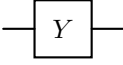
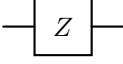
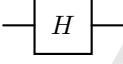
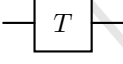
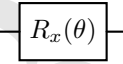
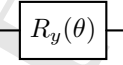
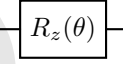
Gate	Circuit	Matrix
X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
H		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
$R_x(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_y(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_z(\theta)$		$\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$

Table 2: Common single qubit quantum gates

for some $|\lambda| = 1$. Hence

$$V = \begin{pmatrix} a & -\lambda\bar{c} \\ c & \lambda\bar{a} \end{pmatrix}. \quad (\text{A.17})$$

Taking determinants gives

$$1 = \det V = \lambda|a|^2 + \lambda|c|^2 \quad (\text{A.18})$$

$$= \lambda(|a|^2 + |c|^2) \quad (\text{A.19})$$

$$= \lambda. \quad (\text{A.20})$$

This also gives $|a|^2 + |c|^2 = 1$. Choose $\gamma, \eta, \mu \in \mathbb{R}$ such that

$$a = e^{i\eta} \cos(\gamma/2), \quad (\text{A.21})$$

$$c = e^{i\mu} \sin(\gamma/2). \quad (\text{A.22})$$

Then

$$V = \begin{pmatrix} e^{i\eta} \cos(\gamma/2) & -e^{-i\mu} \sin(\gamma/2) \\ e^{i\mu} \sin(\gamma/2) & e^{-i\eta} \cos(\gamma/2) \end{pmatrix}. \quad (\text{A.23})$$

On the other hand, direct multiplication gives

$$R_z(\beta)R_x(\gamma)R_z(\delta) = \begin{pmatrix} e^{-i(\beta+\delta)/2} \cos(\gamma/2) & -ie^{-i(\beta-\delta)/2} \sin(\gamma/2) \\ -ie^{i(\beta-\delta)/2} \sin(\gamma/2) & e^{i(\beta+\delta)/2} \cos(\gamma/2) \end{pmatrix}. \quad (\text{A.24})$$

Thus it is enough to choose β, δ so that

$$e^{i\eta} = e^{-i(\beta+\delta)/2}, \quad (\text{A.25})$$

$$e^{i\mu} = -ie^{i(\beta-\delta)/2}. \quad (\text{A.26})$$

Equivalently, modulo 2π , solving gives

$$\beta = \mu - \eta + \frac{\pi}{2}, \quad (\text{A.27})$$

$$\delta = -\mu - \eta - \frac{\pi}{2}. \quad (\text{A.28})$$

Therefore $V = R_z(\beta)R_x(\gamma)R_z(\delta)$. Since $U = e^{i\alpha}V$, we get

$$U = e^{i\alpha}R_z(\beta)R_x(\gamma)R_z(\delta) \quad (\text{A.29})$$

as desired. $\square$

FACT A.13 For matrices A, B, C, D over $\mathbb{C}$, scalar $\lambda \in \mathbb{C}$, and vectors $|\phi\rangle$ and $|\psi\rangle$ over $\mathbb{C}$, the tensor product satisfies:

1. Bi-linearity: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, so $(A \otimes B)(|\phi\rangle \otimes |\psi\rangle) = (A|\phi\rangle) \otimes (B|\psi\rangle)$.
2. Scalars commute: $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$.

3. Associativity: $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.

4. Unitary-preserving: If A, B are unitary matrices, $A \otimes B$ is unitary.

DEFINITION A.14 (MULTI-QUBIT GATES) An n -**qubit gate** is a unitary matrix in $\mathbb{C}^{2^n \times 2^n}$. If gates $U_1, \dots, U_k$ satisfy $U = \bigotimes_{i=1}^k U_i \in \mathbb{C}^{2^n}$, by Fact A.13.4 U is an n -qubit gate. Special named multi-qubit gates are given in Figure 3.

DEFINITION A.15 (CONTROLLED GATES) Many of the gates in Table 3 are **controlled** gates. Informally, controlled gates such as CNOT only apply their target operation when all controls $\text{---}\bullet\text{---}$ are in the state $|1\rangle$. Formally, for any n -qubit gate U , the singly controlled gate, denoted $C-U$, is defined by

$$C-U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (\text{A.30})$$

This definition generalizes naturally to multiple controls.

REMARK A.16 By the above definition, CNOT is just $C-X$, and Toffoli is $CC-X$. Note that we may suppress the hyphen when the meaning is clear.

The gate $C-Z$ is special because either qubit can be interpreted as the control: it only applies a phase when both qubits are in the state $|1\rangle$. To see this, note that

$$I = |0\rangle\langle 0| + |1\rangle\langle 1|, \quad Z = I - 2|1\rangle\langle 1|. \quad (\text{A.31})$$

Therefore,

$$C-Z = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (I - 2|1\rangle\langle 1|) \quad (\text{A.32})$$

$$= I \otimes I - 2|11\rangle\langle 11|. \quad (\text{A.33})$$

Thus $C-Z$ acts as the identity on every computational basis state except $|11\rangle$, where it applies a phase of -1 .

Gate	Circuit	Matrix
CNOT		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
CCNOT or Toffoli		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
C-Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

Table 3: Common multi-qubit quantum gates

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